

Edge4AV: Communication Latency for Connected and Automated Vehicle Applications

Anonymous Author(s)

Abstract

Are today's vehicle-to-everything (V2X) and fifth-generation (5G) communication paths ready for tomorrow's connected and automated vehicle (CAV) applications? Conventional connected-vehicle (CV) and intelligent transportation system (ITS) applications exchange compact, stateless SAE International J2735 messages. Complex CAV applications instead use correlated, stateful operations. These operations may carry larger objects and generate event-triggered bursts or sustained streams in either direction. The current landscape lacks a common interface for both classes. Existing studies do not establish where either path misses application deadlines when carrying both workload classes.

In this paper, we contribute the Intersection Programming Interface (IPI) and a detailed real-vehicle communication-latency study. IPI provides a transport-independent interface for the SAE International J2735 message set and correlated CAV operations. The study evaluates both classes over a commercial Long-Term Evolution cellular V2X (LTE C-V2X) direct PC5 path and a private 5G New Radio (NR) network-assisted Uu path. The 5G deployment uses one physical CAV user equipment (UE), a dedicated radio, and a clean 40-MHz n48 channel without ambient contention. Competing traffic and application-client demand are introduced only as experimental variables. Experiments vary payload size, direction, transport, radio conditions, and interruption. We report response round-trip time (RTT), the percentage of issued requests answered by each deadline, and directional application goodput.

The measurements yield three insights. 1) The measured commercial direct V2X path supports compact J2735 messages only within a limited payload and coverage envelope. 2) The CAV application-packet size that the measured 5G uplink completes within a decision deadline is severely limited. 3) 5G and sixth-generation (6G) radios must support event-triggered bursts and sustained streams in either direction while isolating concurrent flows if cellular networks are to support CAV and ITS applications that advance the goal of zero road fatalities.

1 Introduction

Connected and automated vehicles (CAVs) and connected human-driven vehicles need information that may be visible only to other vehicles or roadside infrastructure. Examples include an occluded road user, the current traffic

signal state, and a cooperative maneuver plan. Vehicle-to-everything (V2X) and fifth-generation (5G) communication can deliver this information to automated driving systems and human drivers [2, 28].

These applications create two communication patterns. Connected-vehicle (CV) and intelligent transportation system (ITS) applications commonly exchange compact, stateless SAE International J2735 updates. A later update replaces an earlier one without preserving an application session. Complex CAV applications instead use correlated, stateful operations for cooperative perception, trajectory planning, maneuver coordination, or remote assistance. An event or fault can trigger an uplink-heavy or downlink-heavy burst, while remote teleoperation can sustain video and control traffic. Both application classes are proposed for direct V2X and network-assisted 5G communication, yet they lack one application protocol and a common evaluation of their communication limits. We therefore ask: *Can today's V2X and 5G paths support both CV and CAV applications, and at what workload size and operating conditions does each path stop meeting the required response latency, completion rate, or application goodput?*

We answer this question with the Intersection Programming Interface (IPI) and a controlled, real-vehicle field study. IPI provides one transport-independent application protocol for stateless CV/ITS messages and stateful CAV operations. The study applies both workload classes to a commercial Long-Term Evolution cellular V2X (LTE C-V2X) direct PC5 path and a private 5G New Radio (NR) Uu path. The private network contains one physical CAV user equipment (UE), a dedicated radio, and a clean 40-MHz n48 channel. Experimental traffic introduces all contention. Across payload, direction, transport, radio condition, configuration, competing traffic, demand, and interruption, we measure response round-trip time (RTT), the percentage of issued requests answered within each application deadline, and directional application goodput.

This paper makes two contributions:

- **A common CV, ITS, and CAV application protocol.** IPI carries typed or pre-encoded J2735 messages and correlated CAV operations through the same transport-independent interface.
- **A controlled communication performance evaluation.** A real autonomous vehicle carries representative CV and CAV workloads over direct PC5 and

network-assisted Uu. The experiments locate the conditions under which each path misses application deadlines or required directional rates.

The measurements yield three insights:

- 1) **The measured commercial direct V2X path supports compact J2735 messages only within a limited payload and coverage envelope.** Roadside-unit placement or a supplemental path is necessary where buildings, highway geometry, or sparse infrastructure leave weak or absent coverage.
- 2) **The CAV application-packet size that the measured 5G uplink can complete within a decision deadline is severely limited.** CAV developers must select, compress, or progressively transmit vehicle-originated state until the uplink envelope expands.
- 3) **5G/6G systems must support event-triggered CAV bursts, sustained directional streams, and concurrent deadline-sensitive traffic.** Radio configurations and schedulers must preserve both directions and isolate simultaneous flows according to application deadline and duration.

2 Related Work

Prior work provides standards for interoperable communication, systems that connect communication to autonomous-driving tasks, and field measurements of direct and network-assisted paths. Figure 1 organizes these foundations around four requirements for cooperative transportation. The comparison exposes two unresolved capabilities: one application contract for CV, ITS, and CAV applications, and comparable measurements that locate the application support boundaries of direct and network-assisted communication.

Standards and application platforms. Vehicular interoperability is divided across message, service, and edge layers. SAE J2735 defines safety, road-user, map, signal, and priority messages for CV and ITS applications [30]. The 3GPP V2X architecture defines PC5 and Uu communication, while its V2X Application Enabler supports registration, delivery, resource management, continuity, and session procedures [1, 3]. The ETSI Multi-access Edge Computing V2X Information Service provides PC5/Uu provisioning, predicted quality of service, and publish/subscribe procedures [15]. Architectures and platforms from the 5G Automotive Association, ETSI, V2X Hub, and CARMA connect selected actors and functions [5, 14, 35, 36].

These layers solve complementary interoperability problems. J2735 defines individual messages, 3GPP manages application services and sessions, and MEC distributes V2X information. Their composition does not provide one implemented application-level data model that preserves J2735

messages and represents the state of a correlated CAV operation through the same interface. IPI provides this common data model and transport-independent interface.

CAV systems and middleware. EMP and AutoCast select and exchange cooperative perception information [29, 41]. VIPS, VI-Map, and SOAR return fused perception, compact maps, and coordinated services from roadside infrastructure [19, 32, 33]. Their objects range from intermediate features to detections, trajectories, and control state. This representation determines the transmitted size, while the driving task determines its deadline.

D3 schedules autonomous-driving operators against end-to-end deadlines [18]. Tentacles selects or combines networks, adapts data quality, and invokes local fallback when a remote result misses its deadline [39]. AutowareV2X selects the fresher collective perception message received over Wi-Fi or LTE [7]. These systems demonstrate application-aware execution and communication, but their exchanged objects and operation state remain application specific. IPI makes the message type, operation identity, timing, and response state available through one interface, allowing network-selection and adaptation mechanisms to operate on both CV and CAV applications.

Direct V2X and cellular field evidence. SEE-V2X combines commercial C-V2X traces with awareness and perception workloads. Ku et al. measure complete fragmented-image delivery, National Institute of Standards and Technology testing examines device interoperability, and recent hardware-in-the-loop and CooperScene studies connect PC5 behavior to control and perception [24, 25, 34, 38, 40]. Together, these studies show why packet delivery alone can overstate application availability: a missing fragment or late update can invalidate an object.

Cellular studies use application-specific completion measures. Vehicle-data upload, teleoperation, and TARGET-X trials report throughput, frame latency, freshness, service quality, or edge behavior [9, 17, 23, 26]. The Next Generation Mobile Networks Alliance identifies implementation and interoperability requirements for balanced and uplink-oriented time-division-duplex frames [27]. Demircioglu’s 40-km private 5G standalone corridor provides the closest cross-path baseline: it integrates PC5 and Uu and measures application-layer latency and packet delivery for six Cooperative ITS scenarios across speed and traffic-density conditions [13].

The literature establishes useful applications, middleware, and deployed PC5 and 5G communication. It does not yet provide a common CV/CAV interface and a cross-path evaluation that applies both compact messages and larger stateful objects, retains every issued request in the denominator, and tests each response against its application deadline. IPI supplies the common interface. The field study then locates PC5 and Uu support boundaries across payload, direction, radio

Toward Zero-Fatality Cooperative Transportation

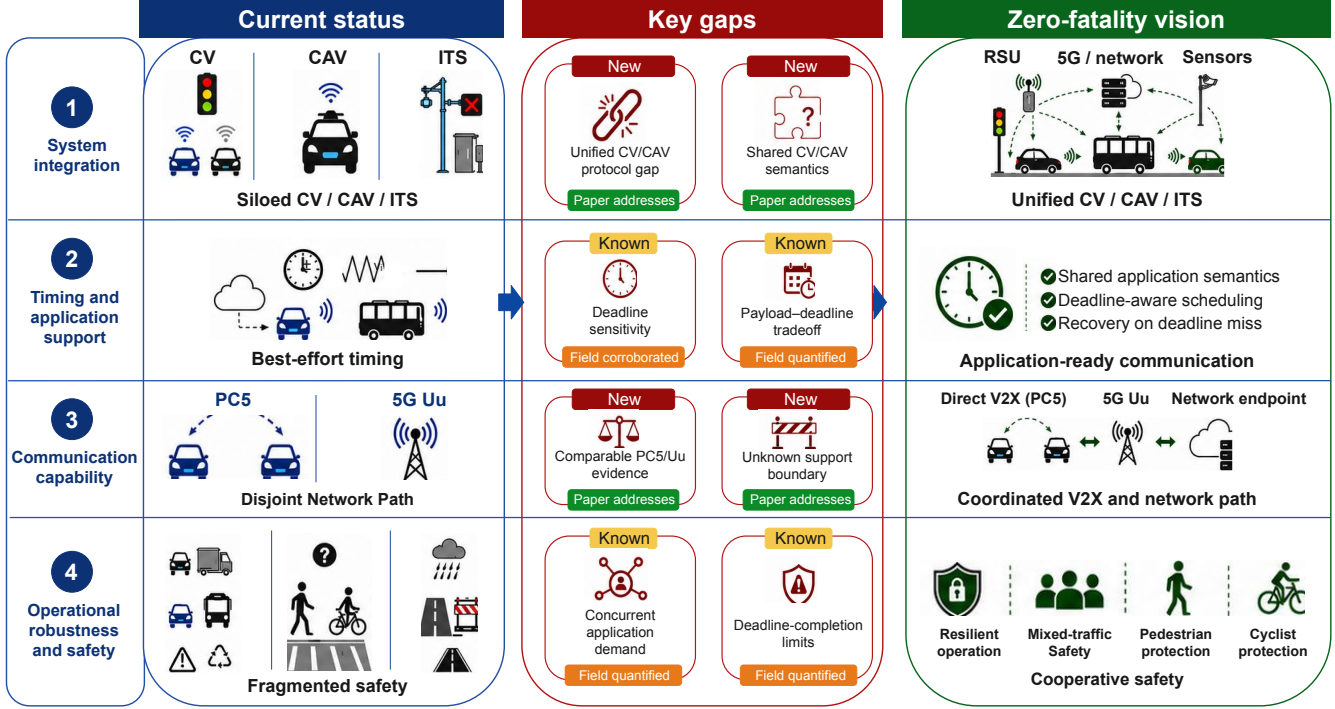


Figure 1: Communication capabilities required for cooperative transportation. Each row connects an existing limitation, the protocol or measurement gap addressed in this paper, and the capability needed for system integration, application timing, path support, or operational safety.

condition, continuous traffic, concurrent demand, and interruption. Section 3 defines IPI, and Sections 4 and 5 present the evaluation.

3 IPI Protocol Design

The preceding literature establishes two requirements for a common application contract: preserve deployed J2735 messages and represent correlated CAV operations. IPI implements both requirements through one transport-independent interface. Stateless CV and ITS exchanges use individual J2735 messages for vehicle state, signal timing, and intersection geometry. Stateful CAV operations use requests, updates, and responses that remain associated through an identity, progress state, and expiration time. Both application classes use the same vehicles and communication resources. IPI is the first implemented application protocol that combines the J2735 message set with correlated stateful CAV operations [3, 30].

Two interaction modes. IPI represents these two application classes through two interaction modes. Message mode carries individual J2735 messages, including stateless updates, through typed helpers for Basic Safety Message

(BSM), Personal Safety Message (PSM), Map Data (MAP), Signal Phase and Timing (SPaT), Signal Request Message (SRM), and Signal Status Message (SSM). It also preserves pre-encoded J2735 bytes and their declared message type. Operation mode carries stateful perception, planning, control, or map operations through registration, heartbeat, request, update, completion, rejection, telemetry, and termination.

Common operation metadata. Both interaction modes use a common envelope. The envelope identifies the message, source, application domain, send time, and selected transport. Session and correlation identifiers bind responses to an operation. Requests add vehicle location, capability, horizon, and application context. Responses add progress, expiration, confidence, and typed or opaque content. The common metadata identifies individual J2735 messages and associates CAV responses with one request and deadline. Figure 2 shows how both interaction modes pass through the common IPI envelope before binding to a communication path and transport.

Reference implementation and validation. The C++17 library implements both modes with message and operation facades, byte encodings, validators, session records, and

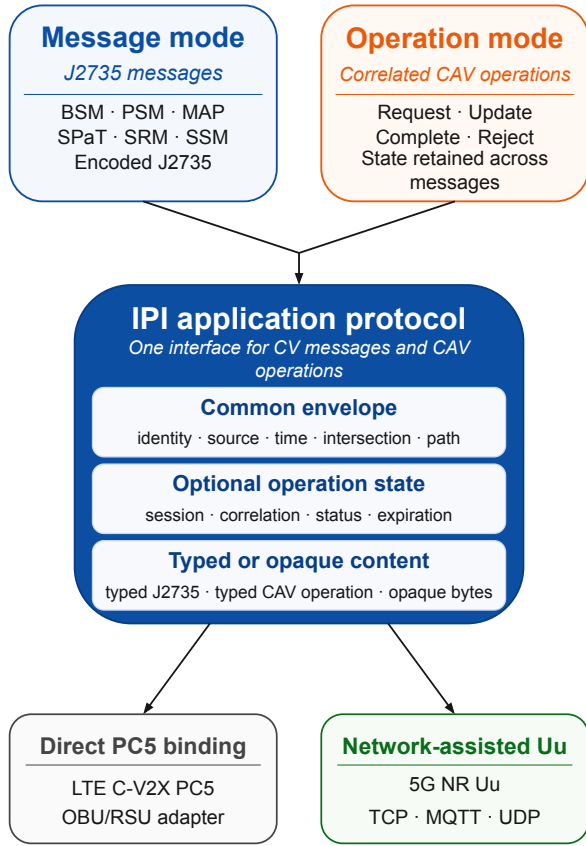


Figure 2: IPI supports stateless CV/ITS applications and stateful CAV applications through one transport-independent application protocol before binding their messages and operations to a communication path and transport.

communication adapters. Typed helpers map common J2735 structures into the IPI envelope. The encoded-payload path preserves the complete J2735 message set. Cooperative requests and responses use the shared envelope, session, and correlation fields. Seven automated tests cover typed and pre-encoded messages, cooperative objects, session operations, malformed input, and Transmission Control Protocol (TCP) and Message Queuing Telemetry Transport (MQTT) loopback. Section 5.1 quantifies representation overhead for preserved J2735 messages and stateful CAV operations.

4 System Design and Experimental Setup

The IPI protocol supports stateless CV/ITS applications and stateful CAV applications. We use workload sizes and directions representative of both communication patterns to compare direct PC5 and network-assisted Uu. We measure

response RTT and deadline completion for each workload. The exact directional transfers additionally measure receiver-observed application goodput after byte-count and checksum validation. The study covers the two access modes emphasized by the 3GPP V2X architecture: direct PC5 communication and network-assisted Uu communication [1]. PC5 exchanges data locally between vehicles and roadside infrastructure. Uu carries vehicle traffic through a radio access network and mobile core to an application endpoint. The experiments use one commercial implementation of each path on the same autonomous vehicle. Protocol fields and serialization additions are reported in bytes (B). Application payloads and transferred objects use kibibytes (KiB = 2^{10} B) or mebibytes (MiB = 2^{20} B). Link rates and application goodput use decimal kilobits per second (kbit/s), megabits per second (Mbit/s), or gigabits per second (Gbit/s).

4.1 Testbed and Measured Paths

Figure 3 shows the physical installation and the radio deployment. The vehicle host generates each workload and records sender-side round-trip time (RTT). A Mocar on-board unit (OBU) provides its PC5 interface, while an MG52 or GL-X3000 gateway provides its Uu interface. At the fixed site, a HUALI roadside unit (RSU) terminates PC5, and an outdoor Airspan AirSpeed 2900 gNodeB terminates Uu. Both paths reach the d1 application host through direct wired connections to the RSU or the on-site MX250. The remote MX68CW manages the private 5G deployment and remains outside the measured user-data path [10–12].

The PC5 equipment uses the vendor’s 2020 J2735 stack. We first exercised its standard-message callbacks. The performance probes then used the vendor’s packet-data interface with a sequence number and same-size echoed payload. The installed interface accepts application packets up to 4,080 B, and the evaluated sweep uses 0–2 KiB. The vendor reports protocol-conformance certification for the product families [21].

Most Uu campaigns connect the vehicle host to a Cisco Meraki MG52 gateway over 1-Gbit/s Ethernet. The MG52 contains the subscriber identity module (SIM) and an internal antenna placed face-up on the trunk floor. The August 19 directional repeat replaces the MG52 with a GLiNet GL-X3000 gateway and evaluates both TDD configurations with that device. Each gateway reaches d1 through the AirSpeed gNodeB, the Cisco private 5G system, and the on-site MX250.

Figure 4 expands these components into the measured request and response paths. A PC5 request returns a sequence-matched, same-size echo. The Uu probes carry IPI Cooperative Service frames. In an uplink-heavy exchange, the vehicle sends the declared application object, and d1 returns a compact correlated acknowledgment after decoding it. In

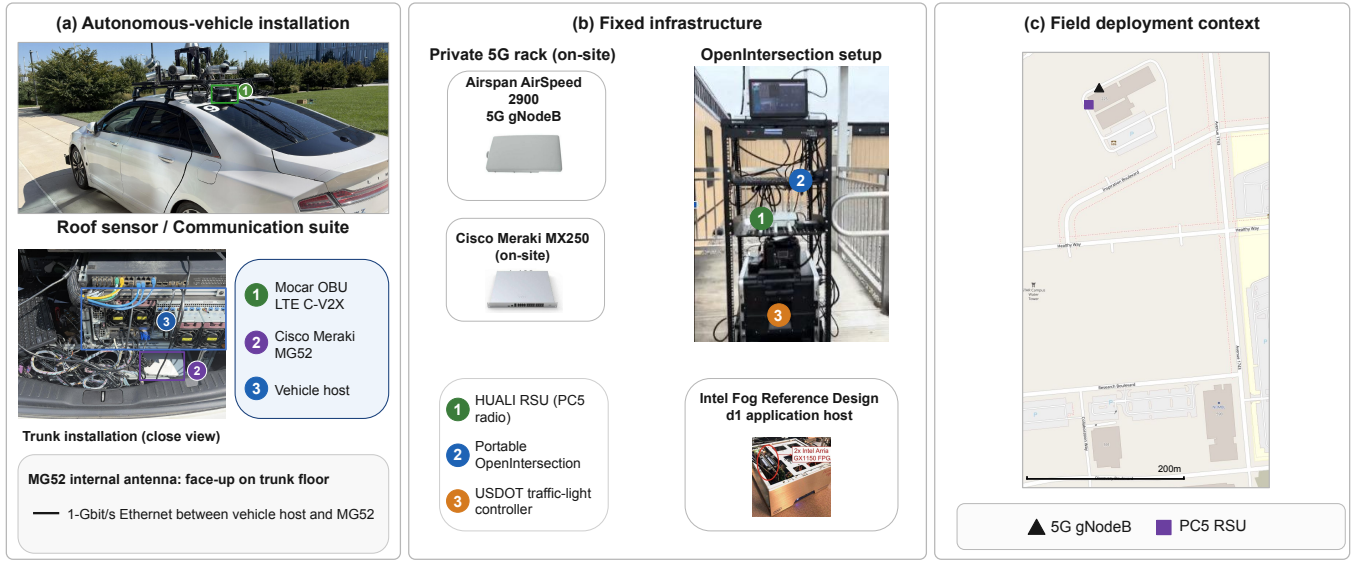


Figure 3: Physical testbed. (a) The autonomous vehicle carries the Mocar OBU, MG52 gateway, and vehicle host. (b) The fixed installation contains the AirSpeed 2900 gNodeB, MX250, HUALI RSU, traffic-light controller, portable OpenIntersection, and d1 application host. (c) The map locates the gNodeB and PC5 RSU. Map data © OpenStreetMap contributors.

a downlink-heavy exchange, the vehicle sends a compact request, and d1 returns the requested application object. The vehicle validates the response sequence number, length, and checksum before stopping its RTT timer.

The Uu deployment was tightly controlled. The vehicle gateway was the sole user equipment (UE), and the AirSpeed radio and its full 40-MHz channel were dedicated to the experiment. The Citizens Broadband Radio Service (CBRS) n48 band was clean and carried no ambient traffic. The baseline measurements contained no unrelated user devices or ambient radio-band contention. Background streams and 1–100 application clients were introduced only as controlled experimental variables behind the same UE attachment. Table 1 summarizes the controls that bound the application measurements and identifies the comparison enabled by each control.

The 5G radio uses 30-kHz subcarrier spacing and TDD. The operator reported 70/20/10 as the deployment’s established allocation. These values denote the downlink, uplink, and dynamic shares. All application campaigns before the TDD study used this profile. Its downlink-oriented allocation is consistent with the downlink-heavy frame structures reported for current 5G deployments [27]. We also evaluate 40/40/20, which assigns 40% of the frame to fixed uplink use rather than 20%. Both profiles retain 10D4G frame packing. The TDD study includes repeated uplink-heavy RTT sweeps, matched downlink-heavy RTT sweeps, and exact 50-MiB transfers in each direction. A replacement-gateway

repeat uses the GL-X3000 for complete uplink-heavy and downlink-heavy matrices and three exact transfers per direction under each profile. Each run records the application direction, location, and available signal context; the earlier runs additionally record available cell state and serving-cell context. The Airspan distributed-unit (DU) Cell exports provide five-minute cell volume, active-time, and unavailability counters for the available windows. These counters verify the serving cell and measurement-block context. Endpoint records define application RTT and goodput.

A Samsung 22 handset running G-NetTrack Pro recorded radio conditions during three continuous drives through the testbed. The combined trace contains 1,301 Global Navigation Satellite System (GNSS) records. We remove 123 records from a frozen radio-value interval longer than 5 s and retain 1,178 valid New Radio (NR) band-48 records. Reference Signal Received Power (RSRP) spans -121 to -88 dBm, with a median of -108 dBm. The corresponding NR signal-to-interference-plus-noise ratio (SINR) spans -20 to 30 dB, with a median of 9 dB. G-NetTrack Pro’s ordinary signal-to-noise ratio field is empty. We recover SINR from the timestamped NR ssSinr entries in the verbose logs and align each entry to its nearest GNSS record.

3GPP defines the 5G Synchronization Signal Reference Signal Received Power (SS-RSRP) measurement but does not assign qualitative coverage levels [4]. We aggregate the outdoor pedestrian levels in Australia’s 2026 coverage-map standard [8]: Good maps to strong ($\text{RSRP} \geq -95$ dBm), Moderate

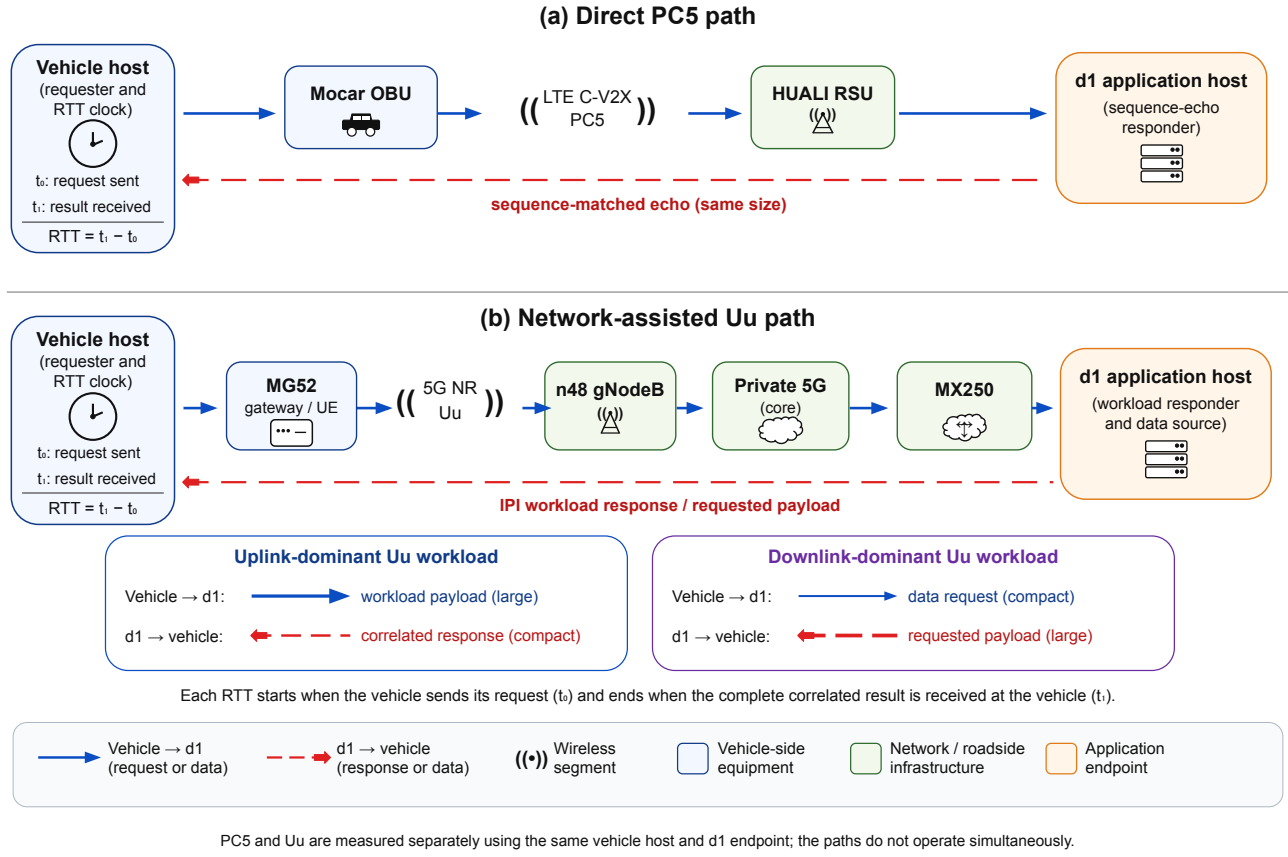


Figure 4: Detailed request/response data paths and sender-side RTT boundaries. (a) The direct PC5 request returns a same-size, sequence-matched echo. (b) The network-assisted Uu path carries either a large vehicle-originated IPI workload followed by a compact correlated response or a compact vehicle request followed by the complete requested payload. PC5 and Uu were measured separately.

Table 1: Experimental controls and their roles in the analysis.

Dimension	Controlled condition	Role in comparison
Radio occupancy	Dedicated 40-MHz n48 channel; no ambient users or traffic	Establishes the single-UE baseline; load is introduced only by the experiment
Client population	One physical CAV UE; 1–100 logical clients share its attachment	Varies application concurrency without changing the number of radio clients
Endpoint and timing	d1 directly connected to the RSU and MX250; unsynchronized endpoint clocks	Measures sender-side RTT and validated goodput; does not infer one-way delay
Field context	Stationary workload records; continuous-drive RSRP/SINR survey; GNSS-joined PC5 outcomes	Relates application measurements to stationary conditions and route coverage
Configuration and gateway	70/20/10 and 40/40/20; MG52 and GL-X3000 repeats	Tests whether TDD behavior persists across vehicle gateways and both traffic directions

maps to common/typical ($-105 \leq \text{RSRP} < -95$ dBm), and Basic and No Coverage map to weak ($\text{RSRP} < -105$ dBm).

The temporal sample distribution contains 83 (7.0%) strong, 340 (28.9%) common/typical, and 755 (64.1%) weak records.

We use the drive survey to characterize route-level coverage and each workload campaign’s signal record to compare stationary application measurements. Figure 5 maps local interpolation of RSRP and SINR within the sampled boundary. The PC5 route analysis joins every issued request and received response to GNSS coordinates, producing spatial response-completion and RTT measurements along each driven path.

Figure 4 defines the request and response events at the RTT boundary. The PC5 RTT covers the vendor interface, two direct-radio traversals, response handling, and correlation. The Uu RTT covers the vehicle host, gateway, radio access, private core, local routing, d1 responder, and return path. The d1 responder decodes each workload frame and immediately returns its response. Its direct connections to the RSU and MX250 make local wired transfer negligible relative to the measured communication paths. The clocks were not synchronized, so all timing uses the sender’s round-trip clock. Each timer begins immediately before the vehicle transmits its request. An uplink-heavy timer ends when the vehicle receives the compact acknowledgment. A downlink-heavy timer ends after the vehicle receives and validates the complete requested payload. The exact 50-MiB transfer experiment uses a separate endpoint timer. The receiver verifies the payload size and checksum before reporting the transfer duration and application goodput.

4.2 Application Conditions and Workloads

Connected-vehicle and CAV applications place different data sizes, dominant directions, durations, and deadlines on the communication path. Table 2 maps six representative applications to the workload probes used to evaluate these requirements [2, 6, 20, 37]. Periodic signal and collision warnings create compact shared updates. For emergency-vehicle signal priority, an authorized vehicle sends an SRM to request priority or preemption, and the infrastructure returns the request status in an SSM [31]. Remote assistance for a disabled or malfunctioning CAV can begin with a compact request and return perception, planning, control, or other data from an edge or help-center endpoint. Remote recovery can also add a continuous vehicle-originated video stream. At 25 mph, the vehicle moves 0.11 m in 10 ms and 1.12 m in 100 ms. Under Federal Highway Administration assumptions, its level-road stopping sight distance is about 46.3 m [16]. A 35-mph adjacent vehicle changes its gap to a 25-mph vehicle by 0.45 m in 100 ms. Remote assistance below 10 km/h moves a vehicle at most 0.278 m in 100 ms and 0.556 m in 200 ms. These applications cannot be represented by one payload, direction, duration, or deadline. Their traffic can be periodic, event triggered, or continuous. Perception, intent, and fault state create vehicle-to-edge demand. A compact vehicle request

can instead return raw sensor data, processed perception, planning data, control data, or other state from the edge. Several operations can also share the same attachment while a continuing stream remains active.

The application conditions define four parts of the workload matrix:

- 1) **Payload and direction.** PC5 spans 0–2 KiB. Uu spans 0–2 MiB using a vehicle-originated object followed by a compact acknowledgment. The downlink-heavy probe returns an exact 1-, 10-, 100-, or 500-KiB object after a compact vehicle request. Separate 50-MiB transfers measure validated goodput in each direction.
- 2) **Application-object size.** The Uu sweep includes 922 V2X-Radar object lists: 19.2 KiB at the median, 23.4 KiB at the 99th percentile, and 24.4 KiB at the maximum. Additional dataset objects span 8 KiB–2 MiB, and a 58.6-KiB object provides a stress condition.
- 3) **Transport.** Matched IPI workloads use TCP, Message Queuing Telemetry Transport (MQTT) over TCP, raw User Datagram Protocol (UDP), and UDP with application-level 1,400-B fragments.
- 4) **Demand, configuration, and recovery.** The experiments add one to four sustained streams, 1–100 logical clients, a requested Quality of Service (QoS) label, two TDD profiles, two vehicle gateways, and a 10-s endpoint or broker interruption.

These factors distinguish payload-size effects from transport, direction, configuration, competing traffic, demand, and interruption.

In the request/response experiments, each application client waits for its response or timeout before issuing the next request. At negligible response time, a 200-ms interval allows at most five requests per second per client. The 100 application clients remain application flows behind one vehicle gateway. The resulting matrix validates the unified interface before comparing payload size, protocol, direction, allocation, demand, and interruption through the same communication metrics. Table 3 orders the comparisons from interface packaging and single-path behavior to shared demand and interruption.

All request/response experiments use the same metrics. The endpoint clocks were not synchronized, so sender-side response RTT is the communication-latency metric. For deadline D , we define the deadline-completion rate as the fraction of issued requests answered within D :

$$C(D) = \frac{N_{\text{responses received within } D}}{N_{\text{issued requests}}}. \quad (1)$$

A PC5 timing cycle models the SRM-to-SSM direction in Table 2. The vehicle sends a request, and the infrastructure returns a status message. The measured packet response is

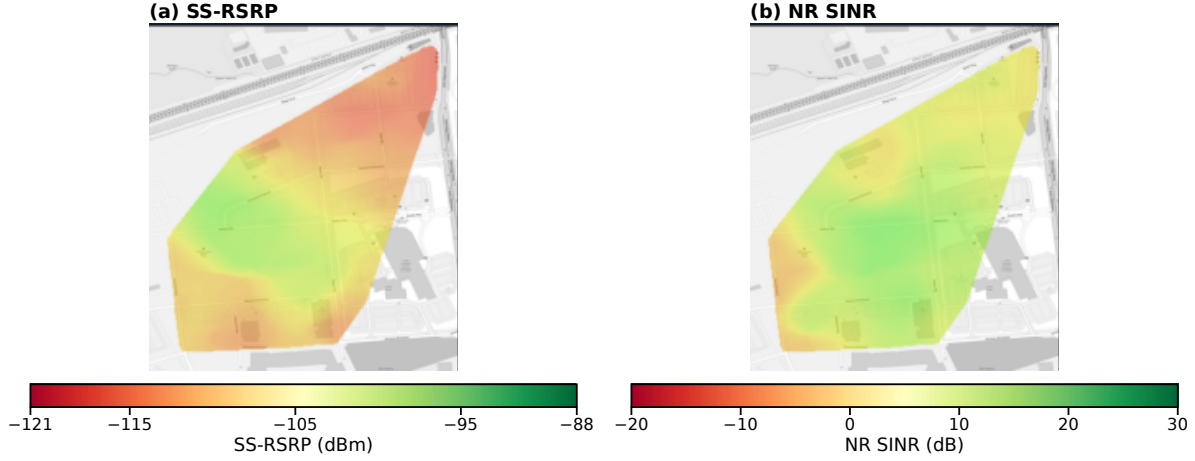


Figure 5: Continuous-drive radio survey of the private 5G testbed. A Samsung 22 handset running G-NetTrack Pro recorded three drive segments. The maps interpolate (a) SS-RSRP and (b) recovered NR SINR from 1,178 valid band-48 records within the sampled boundary. Records from a frozen radio-value interval longer than 5 s are excluded. Map data © OpenStreetMap contributors.

Table 2: Representative application requirements and workload probes. $\uparrow$ denotes vehicle-to-edge traffic and $\downarrow$ denotes edge-to-vehicle traffic. Object sizes exclude transport and security bytes.

Application	Traffic pattern	Object size or rate	Deadline	Workload probe
Signal phase/status	Periodic infrastructure $\downarrow$	About 0.5 KiB at 10 Hz	100 ms	SPaT callback; compact PC5/Uu cycles
Collision or blind-spot warning	Periodic many-to-many PC5	About 0.2 KiB per source at 10 Hz	100 ms	0.25-KiB stationary and route PC5 cycles
Emergency-vehicle signal priority	SRM $\uparrow$; SSM $\downarrow$	0.3–1 KiB	10 ms	SRM/SSM callbacks; 0.25–1-KiB cycles
Emergency maneuver	Reciprocal event-triggered burst $\uparrow\downarrow$	2–12 KiB	3–10 ms	2-KiB PC5; 8–16-KiB Uu cycles
Remote-assistance data request	Compact request $\uparrow$; application object $\downarrow$	1–500 KiB	100–1,000 ms	Downlink-heavy TCP/MQTT cycles
Fault-triggered remote recovery	Video/state $\uparrow$; control/path $\downarrow$	Up to 1 KiB control; 32-Mbit/s video	100/200 ms	Bidirectional Uu; exact transfer, load, demand, and restart

Table 3: Experiment comparisons. $\uparrow$ denotes vehicle-to-edge traffic, and $\downarrow$ denotes edge-to-vehicle traffic.

Experiment	Varied condition	Measured outcome
IPI packaging	J2735 and CAV frames	Added size (B); application coverage
Payload and deadline	PC5 0–2 KiB; Uu 0–2 MiB; 3–1,000 ms	Completion; RTT boundary
Field and route coverage	Stationary points; four PC5 routes	Spatial completion; response tail
Transport	TCP; MQTT; raw and fragmented UDP	Decode/reassembly completion; RTT
Direction	Object $\uparrow$ plus ACK; request plus object $\downarrow$; 50-MiB $\uparrow/\downarrow$	Directional RTT; validated goodput
TDD and gateway	70/20/10 and 40/40/20; MG52 and GL-X3000	Stability; latency; goodput
Shared demand	Sustained streams; 1–100 clients; application label	Foreground completion; achieved rate
Interruption	10-s endpoint restarts	Missing attempts; recovery gap

sequence matched. In an uplink-heavy Uu exchange, the response is a compact acknowledgment returned after the workload frame is decoded. In a downlink-heavy Uu exchange, the response is the complete requested payload, and

its vehicle-side length, sequence, and checksum validation is included in RTT. Every issued request remains in the denominator; missing, rejected, timed-out, and late responses reduce $C(D)$. RTT percentiles describe only the responses

that arrived and are always reported with $C(D)$. We report the median (p50), 95th percentile (p95), and 99th percentile (p99) response RTT. We evaluate D at 3, 10, 25, 100, 200, 500, and 1,000 ms, matching the cooperative, warning, remote-assistance, and relaxed-transfer conditions above. For each application comparison, a measured condition crosses its timing boundary when response p95 exceeds the application deadline. The value $C(D)$ reports how many issued requests satisfy that deadline. The metrics provide different information: RTT characterizes received responses, and $C(D)$ counts every issued request against the deadline.

The exact bulk transfers use receiver-observed application goodput rather than $C(D)$. Each repetition verifies the transferred byte count and checksum, and goodput is the validated payload size divided by receiver-observed transfer duration. The MG52 blocks contain ten repetitions per profile and direction; the GL-X3000 blocks contain three. We report the individual transfers, mean, and a t -based 95% confidence interval. These intervals describe within-block transfer variation.

5 Experiment Results

Each result uses the response RTT and deadline-completion definitions from Section 4 and the application conditions in Table 2. The analysis first measures IPI packaging and direct-PC5 behavior. It then examines 5G uplink payloads, transport, and stationary radio conditions, followed by downlink-heavy workloads and two time-division-duplex (TDD) configurations. The final experiments add sustained streams, concurrent clients, and interruption. This order separates single-path limits from the conditions that place several CAV flows on the same network.

5.1 IPI Packaging Overhead

IPI must carry CV and CAV applications without making message packaging a material part of their communication load. We measure its serialized size addition and verify application coverage.

IPI preserves a pre-encoded J2735 payload of size P B, and the measured message frame adds a 1-B type field and a 4-B length field. Tests cover typed BSM, PSM, MAP, SPaT, SRM, and SSM and byte-exact opaque TIM. The same interface supports stateful CAV requests, updates, completions, and rejections. Table 4 reports the resulting frame sizes before experiment-request and transport headers.

The CAV planning request occupies 63 B without task content. A 256-B content field produces a 334–337-B IPI frame, and the 1,024- and 4,096-B fields produce 1,102–1,105-B and 4,174–4,177-B frames. Thus, IPI adds 78–81 B when task content is present. The added fraction is 30.5–31.6% at 256 B, 7.6–7.9% at 1,024 B, and 1.9–2.0% at 4,096 B. The fixed

Table 4: IPI frame sizes for CV and CAV content.

Content	Application size (B)	IPI frame size (B)	Addition (B)
Preserved J2735 payload	P	$P + 5$	5
CAV planning request	0	63	63
CAV planning and task content	256	334–337	78–81
CAV planning and task content	1,024	1,102–1,105	78–81
CAV planning and task content	4,096	4,174–4,177	78–81

78–81-B addition does not grow with the application object, while the same interface packages the stateless CV/ITS and stateful CAV messages evaluated below.

5.2 Direct PC5 Payload and Route Limits

The packaging experiment bounds the size that IPI adds to each workload. The PC5 experiments measure how payload size and field position affect the resulting exchange across stationary points and driven routes. Figure 6(a) compares response completion as payload and stationary field point change. P1 is building-obstructed at 68 m; P2–P5 are 113, 212, 377, and 469 m from the RSU. The RTT panel uses a separate 7-m reference collection. In that collection, 0.25 KiB returned 999/1,000 responses; 0.5–2 KiB returned every response. Their p50 RTTs remained near 100 ms. Their p95 RTTs ranged from 106.694 to 111.663 ms (Figure 6(b)). These RTTs exceed the 10- and 25-ms cooperative budgets at the median and the 100-ms warning budget at p95.

Payload sensitivity increased across the later field points. The 2-KiB failure rate rose from 0.2% to 99.2%, followed by 910 failures before the next collection was stopped. Over the same sequence, the 1-KiB failure rate rose from 0% to 2.1% and then 82.9%. The 0.25- and 0.5-KiB conditions remained available at more points. The final point returned no response for any payload. The nearby non-line-of-sight (NLOS) point was obstructed by a building and is shown separately from the ordered field sequence.

Route coverage exposes a second limit. The four 0.25-KiB route collections returned 515/675, 731/1,000, 591/1,000, and 716/1,000 responses, yielding 76.3%, 73.1%, 59.1%, and 71.6% completion. Response p95 RTTs were 108.421, 41.887, 43.923, and 41.621 ms. Three routes produced sub-50-ms p95 RTT among received responses, but 26.9–40.9% of their attempts received no response. The GNSS join places the missing responses in recurring weak- and no-path segments. The stationary sweeps establish the payload limit. The route collections show that even compact 0.25-KiB response completion depends on road position and radio-path availability.

5.3 5G Uplink: Payload Size and Deadlines

The PC5 measurements expose payload and route limits for direct exchange. The 5G evaluation extends the payload

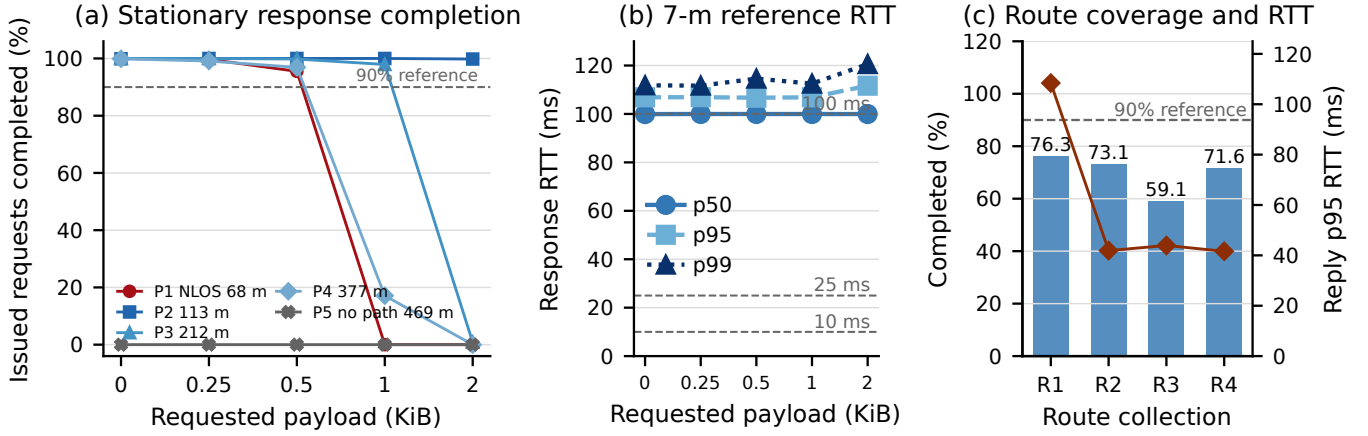


Figure 6: PC5 payload and route results. The response-completion rate is the percentage of issued requests that received a sequence-matched response before timeout. (a) Completion across requested payloads at P1–P5; P1 is the building-obstructed non-line-of-sight point. (b) Response RTT in a separate 7-m reference collection. (c) Completion and p95 RTT for four route collections. The stationary sweep separates payload sensitivity from coverage loss, while the route measurements show that compact requests can still time out along a driven path.

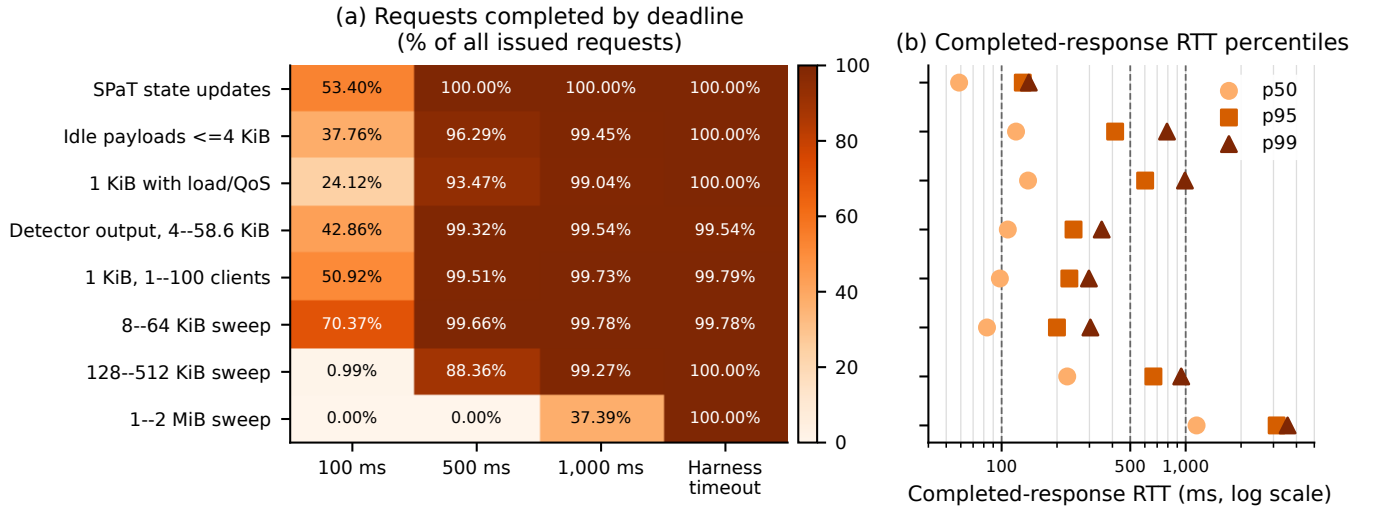


Figure 7: Uplink-heavy private 5G IPI payload-size and deadline results. The vehicle sends the stated workload payload, and d1 returns a compact acknowledgment after decoding it. (a) Percentage of all issued requests completed within 100, 500, and 1,000 ms and before the harness timeout. Detector output denotes serialized object-detection result sizes from 4 to 58.6 KiB replayed as IPI payloads. (b) The p50, p95, and p99 RTT among completed responses.

range over the network-assisted Uu path. Each 5G application probe transmits an IPI cooperative-service workload frame from the vehicle. The d1 endpoint decodes each frame and immediately returns a compact correlated acknowledgment. Figure 7 compares workload responses at 100, 500, and 1,000 ms. Across the eight disjoint workload groups, the analysis contains 752,200 issued requests. Requests with payloads of at most 4 KiB returned all 60,000 responses. Only 37.76% arrived within 100 ms and 96.29% within 500 ms. The

loaded 1-KiB group similarly achieved 24.12% at 100 ms and 93.47% at 500 ms. These conditions carry the payload sizes used by warnings, signal-priority coordination, and compact remote-recovery control. The fraction completed in time decreases as the deadline becomes shorter.

Larger payloads shift the envelope further. The 128–512-KiB group returned all 18,000 responses, with 0.99% within 100 ms and 88.36% within 500 ms. The group with payloads of at least 1 MiB also returned all 14,000 responses. None arrived

within 500 ms, and 37.39% arrived within 1 s. These large vehicle-to-edge workload requests received their compact acknowledgments on timescales that exceed the interactive budgets in Table 2.

Field collections also changed the response tail. Compact-payload p95 RTTs remained within tens or hundreds of milliseconds, while 2-MiB p95 RTTs reached 2.5–10.8 s. A variable-path 256-KiB condition reached a 43.8-s p95. Payload size and field condition jointly shape the usable Uu envelope.

5.4 5G Uplink: Transport Effects on Completion and Latency

Payload size determines how much application data the uplink must carry; transport determines whether the encoded object is delivered as a stream or as one or more datagrams. We therefore compare TCP, MQTT over TCP, raw UDP, and application-fragmented UDP at matched object sizes and field conditions. Figure 8 compares empty-payload controls, detector-derived payloads, and the 58.6-KiB stress payload. TCP and MQTT returned every response at each size in the –106-dBm collection. At –120 dBm, their 58.6-KiB p95 RTTs exceeded 660 ms, and their p99 RTTs exceeded 1 s.

The detector sweep also gives a conservative size boundary for a 500-ms CAV decision budget. Every evaluated TCP and MQTT condition through 19.2 KiB of application content had a p95 RTT below 500 ms in both the –106- and –120-dBm collections. At 22.3 KiB, the –120-dBm MQTT p95 crossed the deadline at 539.780 ms. Later detector points fluctuated rather than restoring a monotonic size guarantee. These measurements place the 500-ms p95 ceiling for this uplink-heavy TCP/MQTT path at approximately 20 KiB of application content before transport framing. The ceiling appeared with one UE on a dedicated 40-MHz n48 channel and no ambient traffic.

Raw UDP completed the empty and 1-KiB conditions but returned no response for the evaluated larger datagrams. Application fragmentation and receiver reassembly sustained at least 93.5% completion through 25 KiB. At 58.6 KiB, completion fell to 20% at –106 dBm and 0% at –120 dBm. TCP and MQTT use retransmission and buffering, while fragmented UDP loses more reassembled responses as payload size and path difficulty grow.

5.5 5G Uplink Across Stationary Radio Conditions

The stationary comparison holds the 70/20/10 profile and uplink-heavy exchange fixed while compact and detector-sized workloads run at four locations. Figure 9 plots the detector-sized results against the reported RSRP at each location. The –120, pooled –109 to –110, and –101 dBm

blocks returned 5,994/6,000, 5,999/6,000, and 6,000/6,000 responses. The larger –109 to –110-dBm campaign returned 307,662/309,000 responses.

Compact 1-KiB exchanges change little within each transport. MQTT p95 spans 40.0–46.1 ms, and 99.1–100% of issued requests complete within 100 ms. TCP p95 spans 137.3–150.4 ms, with 0.5–4.6% completion within 100 ms. The detector-sized workload varies more. MQTT 100-ms completion is 4.4%, 99.4%, 30.8%, and 99.7% at –120, –109 to –110, –106, and –101 dBm. TCP completion is 0%, 86.3%, 18.4%, and 97.8%. Across the same conditions, MQTT p95 RTT is 162.9, 75.8, 183.7, and 59.7 ms; TCP p95 RTT is 188.7, 107.0, 169.5, and 93.6 ms.

The –101-dBm common/typical location produces the lowest detector-sized p95 and the highest 100-ms completion. The –106-dBm condition performs worse than the pooled –109 to –110-dBm conditions, so RSRP alone does not order the four outcomes. Payload size remains stable for the compact exchange and sensitive to the broader radio and path conditions for detector-sized traffic.

5.6 5G Downlink: Returned CAV Objects

A downlink-heavy CAV operation begins with a compact vehicle request and ends after the vehicle receives and validates the returned application object. This pattern represents raw sensor data, processed perception, planning data, control data, or other state requested from an edge application. Under the established 70/20/10 profile, the GL-X3000 downlink campaign returned all 4,000 TCP and all 4,000 MQTT responses across 1, 10, 100, and 500 KiB.

Figure 10 shows p95 RTT rising from 45.8 to 93.4 ms for TCP and from 40.0 to 97.8 ms for MQTT as the returned object grows from 1 to 500 KiB. The 500-KiB response tails remain below the 100-ms reference in this stationary 70/20/10 condition. This established-profile result provides the baseline for testing whether a different radio configuration preserves burst latency and sustained-transfer goodput in both directions.

5.7 5G TDD and Gateway Effects

The TDD study evaluates whether the 40/40/20 behavior persists across vehicle gateways. The MG52 campaigns measured both profiles, but route recovery, serving-cell changes, and weaker signal accompanied part of the 40/40/20 collection. We therefore repeated the complete bidirectional matrix with a GL-X3000 at one stationary location. Within this repeat, TDD is the changing configuration; gateway, location, application, transport, and payload matrix remain fixed.

The GL-X3000 repeat contains 8,000 validated exchanges under each profile. Every exchange returned a response. The

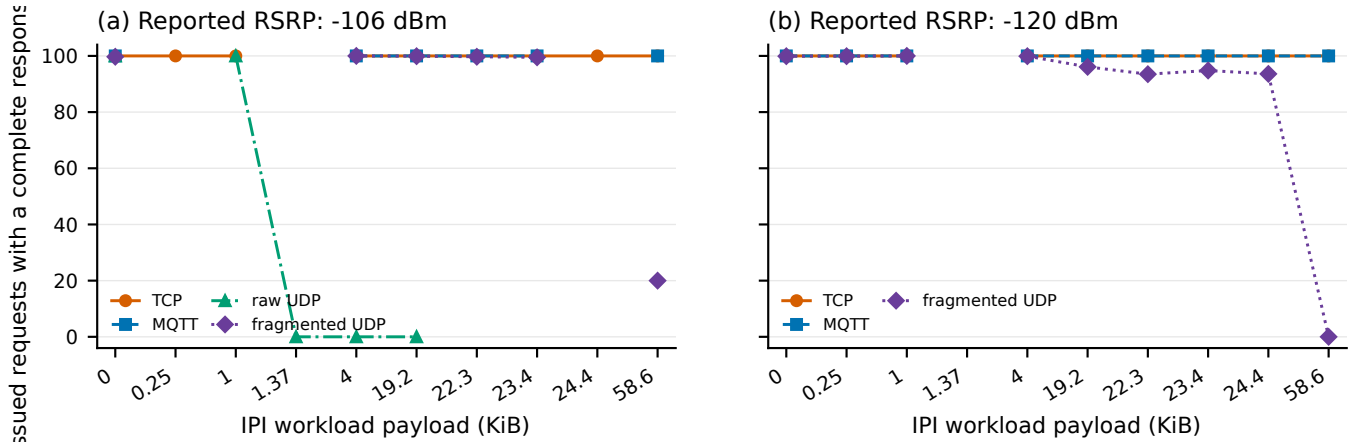


Figure 8: Uplink-heavy response completion by transport, IPI workload size, and reported RSRP. Each line reports the percentage of issued requests that returned a complete response. A gap denotes an unevaluated transport-size pair. Fragmented UDP uses 1,400-B application fragments and completes a response only after reassembly.

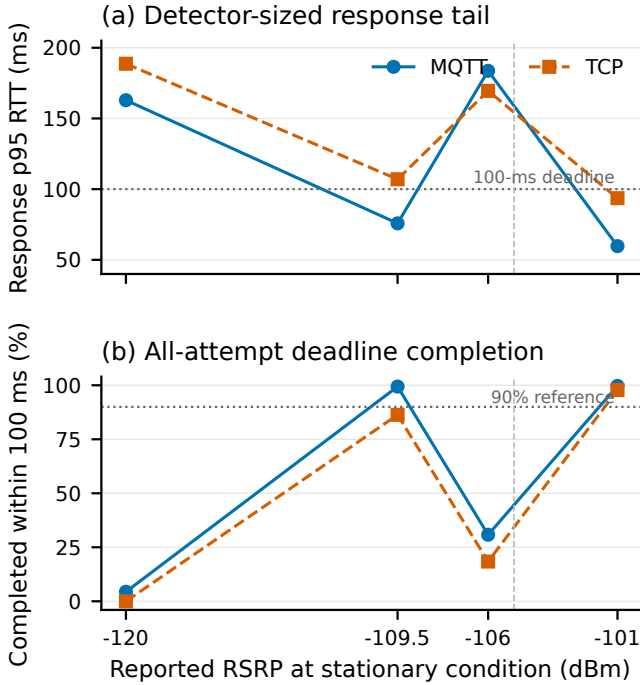


Figure 9: Detector-sized uplink-heavy responses across four stationary 70/20/10 conditions. (a) Response p95 RTT. (b) Percentage of all issued requests completed within 100 ms. The vertical line separates the weak and common/typical RSRP bins defined in Section 4.

profile difference appears in latency rather than eventual completion.

At 1 KiB, vehicle-to-edge p50 RTT was effectively tied. The difference grew with the vehicle-originated object. At

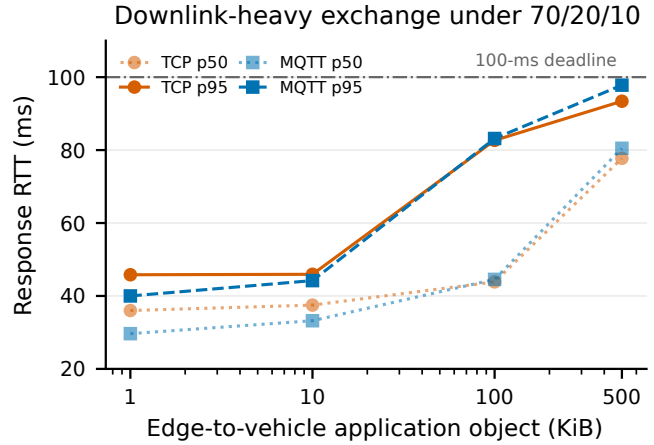


Figure 10: Downlink-heavy CAV object latency under the established 70/20/10 profile. The vehicle sends a compact request and validates the complete 1–500-KiB edge-to-vehicle object. Each point summarizes 500 exchanges.

500 KiB, 40/40/20 increased TCP p50 RTT from 231.046 to 609.652 ms. TCP p95 increased from 258.200 to 1,149.368 ms. MQTT p50 increased from 238.607 to 580.942 ms, while p95 increased from 267.336 to 1,104.331 ms. The corresponding p95 ratios were 4.45 and 4.13. Edge-to-vehicle objects also had higher p95 RTT under 40/40/20 at every evaluated size and transport. The 500-KiB ratios were 3.44 for TCP and 1.41 for MQTT (Figure 11(a–b)).

These tail changes reduced deadline completion across the equally sized workload matrix. Under 70/20/10, 14.54%, 0.04%, and 0% of the 8,000 exchanges exceeded 100, 500,

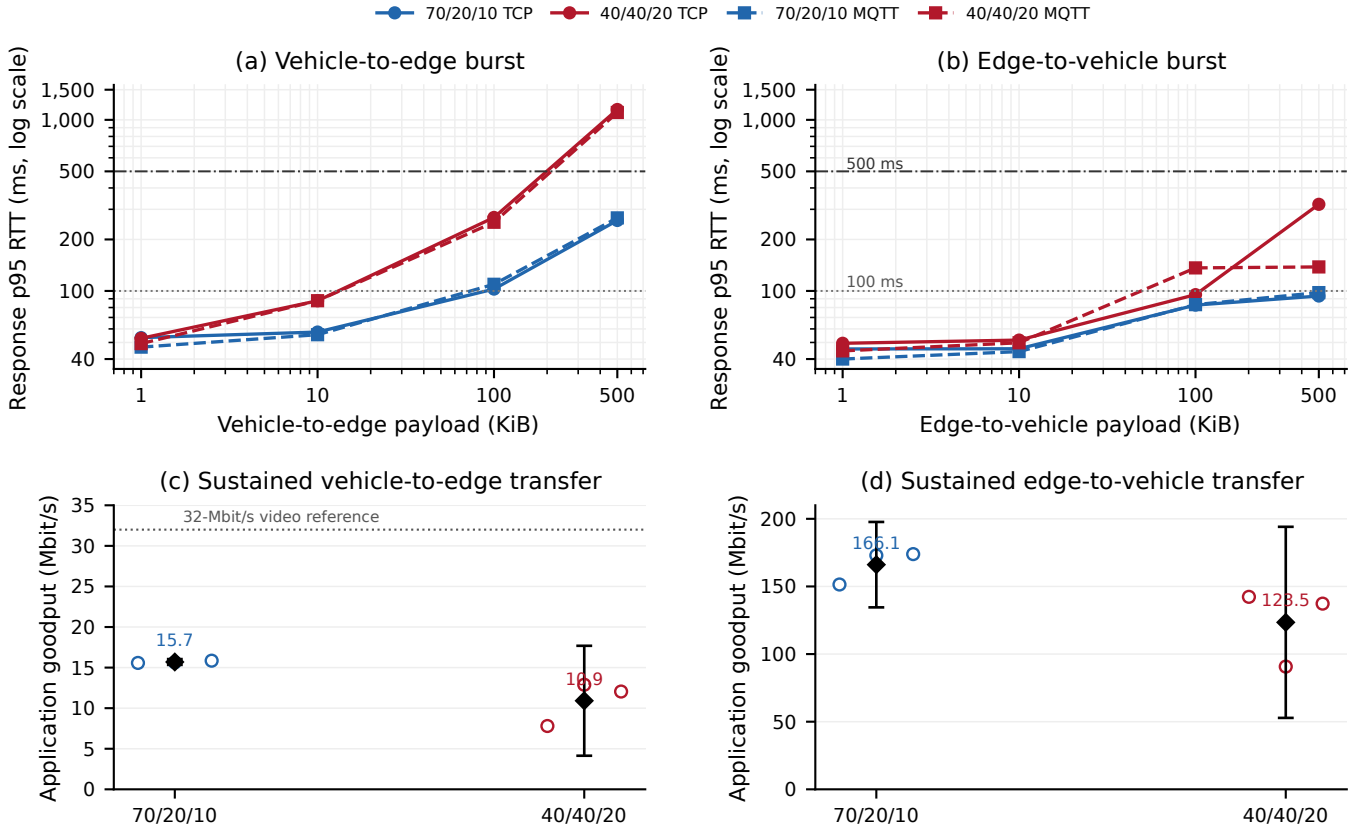


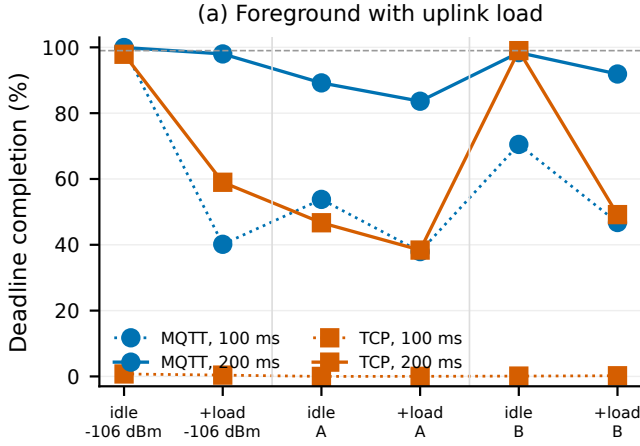
Figure 11: Same-device performance for CAV bursts and sustained transfers. The GL-X3000 blocks use the same stationary location; 70/20/10 and 40/40/20 have median RSRP of -99 and -98 dBm and median SINR of 20 dB. Panels (a–b) combine TCP and MQTT p95 RTT; each profile-transport marker summarizes 500 validated exchanges. The dotted and dash-dotted lines mark 100- and 500-ms application references. Panels (c–d) show three exact 50-MiB transfers per profile and direction; open circles are individual transfers, diamonds are means, and whiskers are t -based 95% confidence intervals. The dotted line in (c) marks the 32-Mbit/s remote-teleoperation video reference.

and 1,000 ms. Under 40/40/20, the corresponding fractions were 30.85%, 9.81%, and 1.26%. The latter profile produced 101 second-scale exchanges; 82 occurred in the two 500-KiB vehicle-to-edge conditions. Thus, the alternative configuration widened the large-burst tail even though it nominally assigns twice as much of the frame to fixed uplink use.

The radio measurements place both profile blocks in the same signal category. Median RSRP was -99 dBm under 70/20/10 and -98 dBm under 40/40/20. Median SINR was 20 dB in both blocks. Restricting the analysis to their common RSRP/SINR range retained 15,153 exchanges and the same latency ordering in 14 of 16 p50 comparisons and 15 of 16 p95 comparisons. At 500 KiB, TCP vehicle-to-edge server processing remained 1.617–1.619 ms at p50 while response RTT changed by 378.606 ms. The latency ordering therefore persists within common measured signal support, while the endpoint processing time remains nearly identical.

The exact transfers measure the sustained-stream component of CAV traffic. Each profile completed three 50-MiB uploads and three 50-MiB downloads with matching byte counts and checksums. Vehicle-to-edge goodput averaged 15.691 Mbit/s under 70/20/10. It averaged 10.913 Mbit/s under 40/40/20, a 30.4% reduction. Edge-to-vehicle goodput averaged 166.106 and 123.451 Mbit/s, a 25.7% reduction. Every 40/40/20 transfer was slower than every corresponding 70/20/10 transfer (Figure 11(c–d)). The two upload means provide 49.0% and 34.1% of the 32-Mbit/s remote-teleoperation video reference in Table 2.

The cross-gateway controls separate device improvement from the profile ordering. Under 70/20/10, GL-X3000 upload and download goodput exceeded the closest MG52 control by 30.8% and 31.0%. The earlier MG52 large-uplink collapse did not recur with the GL-X3000, but the same-device comparison still favored 70/20/10 for burst latency and sustained



(b) Requested and observed packet marking

Requested label default / 5qi-mapped
Observed IPv4 TOS 0x0 / 0x0

Figure 12: Compact foreground exchanges during a sustained uplink stream. (a) Percentage of issued 1-KiB requests completed within 100 and 200 ms at idle and with the uplink generator active; Collections A and B are separate weak-signal field collections. (b) Requested application labels and the observed vehicle-side Internet Protocol version 4 Type of Service (TOS) value.

goodput. Earlier profile transitions also changed route, cell state, and serving cell. Later matched blocks completed their exchanges but retained configuration-dependent latency and goodput. Successful attachment and eventual completion therefore did not ensure predictable deadline performance.

5.8 Concurrent Bursts and Sustained Streams

During remote teleoperation, a vehicle can sustain a video stream while faults, control updates, or planning requests create additional deadline-sensitive exchanges. The load experiment models this communication pattern with a continuing vehicle-to-edge stream and a compact 1-KiB foreground exchange on the same host and UE. With the generator active, MQTT $C(100\text{ ms})$ fell from 99.70% to 40.15%, while TCP $C(200\text{ ms})$ fell from 97.90% to 59.00%. Across the load collections, achieved background traffic ranged from 1.320 to 5.743 Mbit/s. The active generator reduced foreground deadline completion before reaching the offered rate.

The response tails show the same shift. In the -106-dBm repetitions, the active generator increased MQTT p95 from 52–60 to 174–176 ms and TCP p95 from 162–194 to 388–420 ms. The response tail did not vary monotonically with the number of streams, so achieved background rate alone does not predict foreground latency.

The default and 5qi-mapped labels both produced observed TOS values of 0x0 in the vehicle capture (Figure 12(b)). The foreground and background flows entered the measured path without an observed packet-class difference between the requested labels. The sustained stream therefore displaced a compact exchange from its deadline before reaching the 32-Mbit/s video reference, and the requested application label did not create observable packet differentiation on the vehicle interface.

5.9 Concurrent Application Demand

The preceding experiment combines one continuing stream with one foreground exchange. CAVs can also start several perception, planning, and control operations at the same time. The client-demand experiment increases the number of simultaneous application flows behind the same physical UE.

The client-demand experiment scales a homogeneous 1-KiB exchange across multiple application clients on one host and UE. At 100 clients and -106 dBm , aggregate p95 RTT was 81.850 ms for MQTT, 264.070 ms for TCP, and 113.391 ms for UDP. At -120 dBm , MQTT and TCP p95 rose to 5,135.984 and 2,908.420 ms. UDP p95 was 172.500 ms, but 9,189 of 100,000 cycles received no response. Figure 13(c–d) shows that deadline completion falls as client demand grows. Per-client tail distributions further show that individual flows can miss the 100- and 200-ms budgets even when most aggregate requests eventually receive a response.

5.10 Endpoint Interruption and Recovery

The interruption experiment restarted a receiver or broker for 10 s while 1-KiB requests continued. MQTT broker, MQTT receiver, TCP receiver, and UDP receiver restarts returned 955/1,000, 991/1,000, 953/1,000, and 969/1,000 responses. At the 200-ms remote-assistance budget, completion was 94.7%, 97.7%, 92.9%, and 96.9%. The gaps between received responses were 15.1007, 13.3674, 11.7147, and 13.4635 s. Every response gap exceeds the injected outage and is absent from response-only latency percentiles. Among the responses that arrived, the corresponding p99 RTTs were 90.553, 259.906, 361.228, and 63.027 ms.

5.11 Cross-Application Comparison

The preceding subsections report path-specific response latency and deadline completion. Figure 14 places each selected

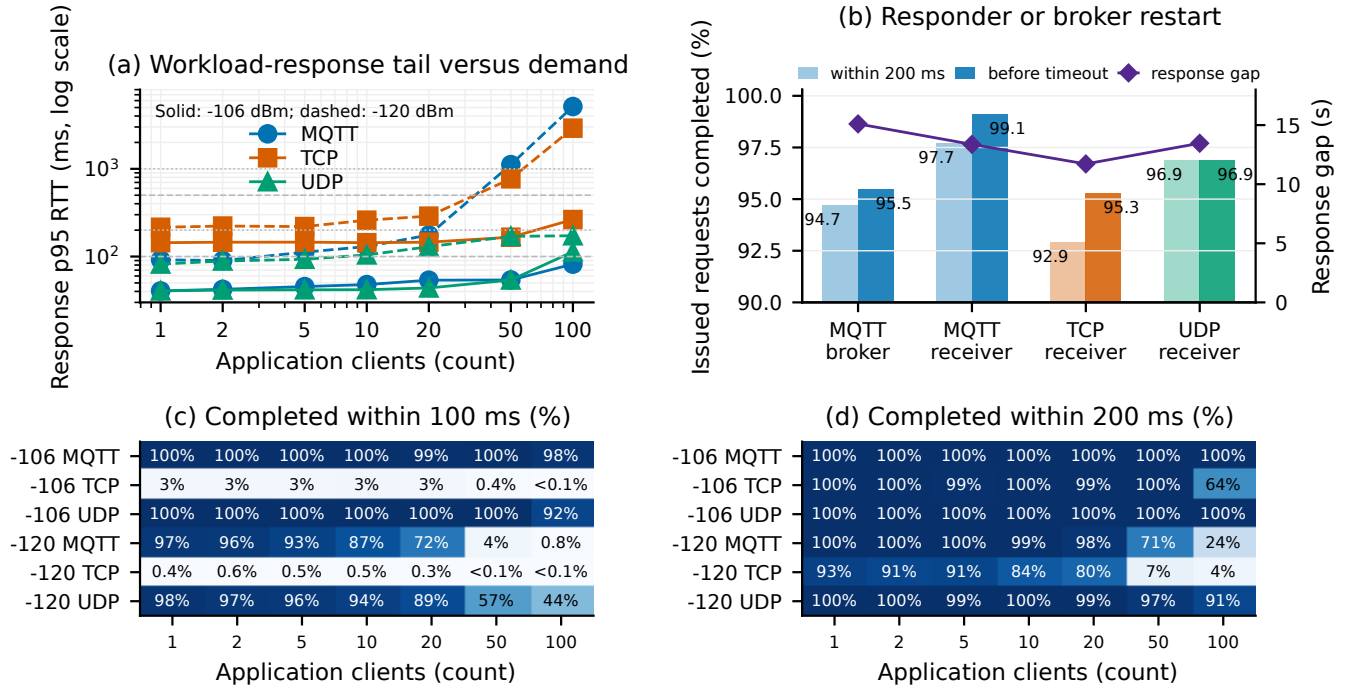


Figure 13: Client-demand and interruption results. (a) Response p95 RTT in milliseconds versus client count; solid and dashed lines mark the -106- and -120-dBm weak-band collections. (b) Completion percentage within 200 ms and before the harness timeout, together with the response gap in seconds after a 10-s restart. (c-d) Percentages of all issued requests completed within 100 and 200 ms.

response tail and its application deadline on the same time axis. A measured marker to the left of its deadline marker meets the p95 timing requirement; a marker to the right misses it.

The 5G MQTT SPaT exchange has a 34.1-ms p95 against the 100-ms signal-warning deadline. The PC5 0.5-KiB response tail reaches 106.9 ms and misses the same deadline. The four PC5 route p95 values span 41.6–108.4 ms, crossing the 100-ms blind-spot-warning deadline. The signal-priority and emergency-maneuver exchanges have 44.5–111.7-ms p95 values against 10-ms deadlines. The loaded remote-recovery control reaches 175.0 ms against its 100-ms deadline. Response completion supplies the corresponding availability: Figure 6(c) shows that only 59.1–76.3% of route attempts receive a response.

Operating conditions also alter the same workload when payload and deadline are held fixed. For the detector-sized workload under 70/20/10, 100-ms MQTT completion is 4.4% at -120 dBm. It reaches 99.7% at the common/typical location with -101 dBm recorded RSRP. The -106-dBm 70/20/10 condition reaches 30.8%. Thus, the same profile and workload cross the application deadline as location changes.

The measured deadline outcome also differed between the two deployed configurations within the same signal category.

In the GL-X3000 comparison, the fraction of all exchanges exceeding 500 ms increased from 0.04% under 70/20/10 to 9.81% under 40/40/20. The latter configuration also reduced sustained upload and download goodput by 30.4% and 25.7%. The corresponding RSRP medians differed by 1 dB, and both blocks had 20-dB median SINR.

Shared demand changes the same deadline a second way. Compact 1-KiB MQTT completion falls from 99.7% to 40.15% with competing uplink traffic (Figure 12(a)) and from 96.6% to 0.751% as -120-dBm client demand grew from one to 100 (Figure 13(c)). The default and 5qi-mapped labels both produced vehicle-side TOS 0x0. Consequently, the measured path exposed no packet-class distinction between these labels as competing traffic and concurrent application demand moved the same 1-KiB workload across its deadline.

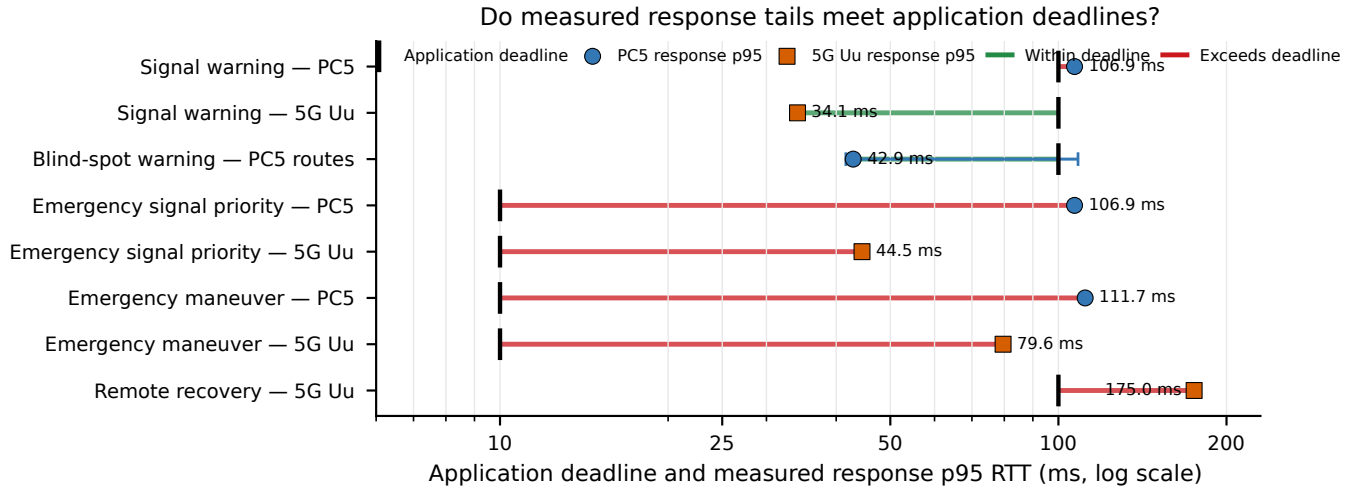


Figure 14: Application deadlines and measured response tails. Each black tick is the deadline from Table 2; each circle or square is the response p95 RTT for the selected PC5 or uplink-heavy 5G condition. Green connectors meet the deadline, and red connectors miss it. The PC5 blind-spot point is the median p95 across four routes, with the interval spanning their p95 values.

5.12 Summary and Insights

The cross-application comparison identifies different payload, coverage, direction, radio-configuration, and demand limits. CAV traffic can begin with an event-triggered uplink or downlink burst and then continue as a sustained stream during remote teleoperation or continuing sensor delivery. Other deadline-sensitive operations can begin while that stream remains active. Increasing the workload and varying the operating conditions reveal where each path crosses the application deadline and how many issued requests remain within it. These measured boundaries lead to three insights:

- 1) **The measured commercial direct V2X path supports compact J2735 messages only within a limited payload and coverage envelope.** The functional tests delivered standard SPaT and BSM messages, and the reference stationary point carried the compact payload sweep. The building-obstructed point at 68 m lost every 1- and 2-KiB exchange. The 113-m point returned 99.8–100% of every tested size. The four route collections lost 23.7–40.9% of compact 0.25-KiB requests. This contrast shows that direct V2X usefulness depends on roadside infrastructure geometry rather than distance alone. Urban intersections can exploit nearby infrastructure. High-rise street canyons, highways, and sparsely served roads may require carefully placed roadside units or a supplemental path.
- 2) **The CAV application-packet size that the measured 5G uplink can complete within a decision deadline is severely limited.** Every evaluated TCP and MQTT detector condition through 19.2 KiB kept

p95 RTT below 500 ms at both measured placements, but the -120 -dBm MQTT collection crossed that deadline at 22.3 KiB. These measurements place the conservative 500-ms p95 ceiling for this path at approximately 20 KiB of application content. For the tighter 100-ms budget, even the aggregate group at or below 4 KiB completed only 37.76% of issued requests in time. These limits emerged on a dedicated 40-MHz channel with one UE and no ambient contention. At the matched 500-KiB, -115 -dBm condition, the downlink-heavy p95 was 226.787 ms while the uplink-heavy p95 was 9,287.828 ms. CAV developers must therefore select, compress, or progressively transmit vehicle-originated state according to the uplink deadline instead of applying one packet-size rule to both directions.

- 3) **5G/6G systems must support event-triggered CAV bursts, sustained directional streams, and concurrent deadline-sensitive traffic.** With one GL-X3000 at the same stationary location and with comparable measured RSRP and SINR, the deployed 40/40/20 configuration produced more deadline misses and lower sustained goodput than 70/20/10 despite its larger nominal uplink allocation. The GL-X3000 repeat measured higher goodput than the closest MG52 control, yet the configuration gap remained. A sustained uplink stream also moved compact foreground exchanges across their deadlines. Radio vendors and mobile network operators therefore need stable configurations across radios

and vehicle gateways. Their schedulers must use application direction, deadline, and duration while isolating simultaneous flows.

6 Future Research Directions

The PC5 results create a coverage-planning problem for connected-vehicle and ITS deployments. Future work should determine roadside-unit spacing, antenna geometry, and supplemental-path placement from the fraction of compact J2735 messages completed within the application deadline. The evaluation must include urban intersections, high-rise street canyons, highways, and sparsely served roads because distance alone does not characterize obstruction or coverage.

The 5G results create a directional representation problem for CAV developers. At the matched large-object condition, vehicle-originated content required substantially longer than the same-sized edge-returned content. Future CAV communication should therefore select, compress, partition, or progressively transmit vehicle-originated state according to the available uplink, the current radio condition, and the decision deadline. These mechanisms also need explicit expiration and recovery so that delayed fragments do not consume resources after the application can no longer use the complete object.

For mobile network operators and 5G/6G researchers, CAV demand changes in arrival time, dominant direction, and duration. A driving event can create an immediate uplink- or downlink-heavy burst. Remote teleoperation can then sustain video, state, and control traffic for the remainder of the assistance session. Other deadline-sensitive operations may begin while this stream remains active. The same-device measurements show that a nominal TDD ratio does not determine end-to-end burst latency or sustained goodput. The load experiments also show that a continuing stream can move compact foreground exchanges across their deadlines. Future radio work should make balanced and uplink-oriented frame structures interoperable across chipsets, vehicle gateways, and infrastructure while preserving attachment and service state. Schedulers must consider each workload's direction, deadline, and expected duration while isolating simultaneous traffic. Common benchmarks should combine asynchronous bursts with sustained streams across independently scheduled UEs while varying signal condition, TDD allocation, mobility, and handover. They should report direction-separated RTT, deadline completion over all issued requests, upload and download application goodput, fairness, configuration-state provenance, and recovery gaps [22, 27].

7 Conclusion

IPI is a transport-independent application protocol for stateless CV/ITS applications and stateful CAV applications. Its

common interface allows both workload classes to be evaluated over direct PC5 and network-assisted Uu rather than assigning each class to one communication technology in advance. In a tightly controlled real-vehicle deployment, the evaluation increases application payload, reverses the direction of large CAV objects, combines finite exchanges with sustained streams, and varies communication conditions. It measures response latency, deadline completion, and exact directional application goodput.

The results establish that the communication envelope is specific to each application. Direct PC5 carries compact J2735 messages when roadside coverage is available, while obstruction and sparse infrastructure reduce their availability. The 5G path carries larger stateful exchanges, but the measured uplink crosses a 500-ms decision deadline near 20 KiB. The GL-X3000 repeat measured approximately 31% higher sustained goodput than the closest MG52 control under the established profile. With the GL-X3000 held constant, the deployed 40/40/20 configuration still produced more deadline misses and lower goodput than 70/20/10 despite its larger nominal uplink allocation. A continuing uplink stream also moved compact foreground exchanges across their deadlines.

These results connect the two contributions. IPI expresses periodic CV updates, stateful CAV bursts, and continuing CAV sessions through one application interface. The communication study then identifies where payload, direction, coverage, configuration, and simultaneous traffic prevent those workloads from meeting their requirements. Supporting CAV applications therefore requires radio vendors and mobile network operators to provide stable bidirectional configurations and scheduling for both event-triggered bursts and sustained deadline-sensitive streams.

References

- [1] 3rd Generation Partnership Project. 2024. *Architecture Enhancements for 5G System to Support Vehicle-to-Everything Services*. Technical Report ETSI TS 123 287 V18.4.0, 3GPP TS 23.287 Release 18. European Telecommunications Standards Institute. https://www.etsi.org/deliver/etsi_ts/123200_123299/123287/18.04.00_60/ts_123287v180400p.pdf
- [2] 3rd Generation Partnership Project. 2025. *Service Requirements for Enhanced Vehicle-to-Everything Scenarios*. Technical Report ETSI TS 122 186 V19.0.0, 3GPP TS 22.186 Release 19. European Telecommunications Standards Institute. https://www.etsi.org/deliver/etsi_TS/122100_122199/122186/19.00.00_60/ts_122186v190000p.pdf
- [3] 3rd Generation Partnership Project. 2026. *Application Layer Support for Vehicle-to-Everything Services: Functional Architecture and Information Flows*. Technical Report ETSI TS 123 286 V18.7.0, 3GPP TS 23.286 Release 18. European Telecommunications Standards Institute. https://www.etsi.org/deliver/etsi_TS/123200_123299/123286/18.07.00_60/ts_123286v180700p.pdf
- [4] 3rd Generation Partnership Project. 2026. *NR: Physical Layer Measurements*. Technical Report ETSI TS 138 215 V19.2.0, 3GPP TS 38.215 Release 19. European Telecommunications Standards Institute. https://www.etsi.org/deliver/etsi_ts/138200_138299/138215/19.02.00_60/ts_138215v190200p.pdf
- [5] 5G Automotive Association. 2020. *V2X Application Layer Reference Architecture*. Technical Report 5GAA A-200094, Version 1.0. 5G Automotive Association. https://5gaa.org/content/uploads/2020/06/5GAA_A-200094_V2XSRA-Application-Layer-Reference-Architecture-final.pdf
- [6] 5G Automotive Association. 2021. *C-V2X Use Cases and Service Level Requirements, Volume II*. Technical Report Version 1.0. 5G Automotive Association. https://5gaa.org/content/uploads/2021/01/5GAA_T-200116_TR_C-V2X_Use_Cases_and_Service_Level_Requirements_Vol_II_V2.1.pdf
- [7] Yu Asabe, Ehsan Javanmardi, Jin Nakazato, Manabu Tsukada, and Hiroshi Esaki. 2024. Enhancing Reliability in Infrastructure-Based Collective Perception: A Dual-Channel Hybrid Delivery Approach With Real-Time Monitoring. *IEEE Open Journal of Vehicular Technology* 5 (2024), 1124–1138. doi:10.1109/OJVT.2024.3443877
- [8] Australian Communications and Media Authority. 2026. *Telecommunications (Mobile Network Coverage Maps) Industry Standard 2026*. Technical Report F2026L00381. Australian Government, Federal Register of Legislation. <https://www.legislation.gov.au/F2026L00381/asmade/text>
- [9] Jason Carpenter, Wei Ye, Feng Qian, and Zhi-Li Zhang. 2023. Multi-Modal Vehicle Data Delivery via Commercial 5G Mobile Networks: An Initial Study. In *2023 IEEE 43rd International Conference on Distributed Computing Systems Workshops*. IEEE, 157–162. doi:10.1109/ICDCSW60045.2023.00026
- [10] Cisco Meraki. 2026. *MX250 Datasheet*. Accessed August 12, 2026. https://documentation.meraki.com/SASE_and_SD-WAN/MX/Product_Information/Overviews_and_Datasheets/MX250_Datasheet
- [11] Cisco Meraki. 2026. *MX67, MX68, and MX68CW Datasheet*. Accessed August 12, 2026. https://documentation.meraki.com/SASE_and_SD-WAN/MX/Product_Information/Overviews_and_Datasheets/MX67_and_MX68_Datasheet
- [12] Cisco Systems. 2025. *Cisco Private 5G Security*. Technical Report. Cisco Systems. <https://www.cisco.com/c/en/us/products/collateral/wireless/cisco-private-5g-solution-wp.html>
- [13] Erdem Demircioglu. 2026. Field Deployment and Performance Evaluation of an NR-V2X C-ITS Test Corridor over a 5G SA Private Network. *Electronics* 15, 12 (2026), 2668. doi:10.3390/electronics15122668
- [14] European Telecommunications Standards Institute. 2010. *Intelligent Transport Systems: Communications Architecture*. Technical Report ETSI EN 302 665 V1.1.0. European Telecommunications Standards Institute. https://www.etsi.org/deliver/etsi_en/302600_302699/302665/01.01.00_30/en_302665v010100v.pdf
- [15] European Telecommunications Standards Institute. 2024. *Multi-access Edge Computing: V2X Information Services API*. Technical Report ETSI GS MEC 030 V3.2.1. European Telecommunications Standards Institute. https://www.etsi.org/deliver/etsi_gs/MEC/001_099/030/03.02.01_60/gs_mec030v030201p.pdf
- [16] Federal Highway Administration. 2009. *Speed Concepts: Informational Guide*. Technical Report FHWA-SA-10-001. U.S. Department of Transportation. <https://highways.dot.gov/safety/speed-management/speed-concepts-informational-guide/chapter-4-engineering-and-technical>
- [17] Rostand A. K. Fezeu, Jason Carpenter, Rushikesh Zende, Sree Ganesh Lalitaditya Divakarla, Nitin Varyani, Faaqi Bilal, Steven Sleder, Nanditha Naik, Duncan Joly, Eman Ramadan, Ajay Kumar Gurumadaiah, and Zhi-Li Zhang. 2025. Teleoperating Autonomous Vehicles over Commercial 5G Networks: Are We There Yet? *arXiv preprint arXiv:2507.20438* (2025). doi:10.48550/arXiv.2507.20438
- [18] Ionel Gog, Sukrit Kalra, Peter Schaffhalter, Joseph E. Gonzalez, and Ion Stoica. 2022. D3: A Dynamic Deadline-Driven Approach for Building Autonomous Vehicles. In *Proceedings of the Seventeenth European Conference on Computer Systems*. ACM, 453–471. doi:10.1145/3492321.3519576
- [19] Yuze He, Chen Bian, Jingfei Xia, Shuyao Shi, Zhenyu Yan, Qun Song, and Guoliang Xing. 2023. VI-Map: Infrastructure-Assisted Real-Time HD Mapping for Autonomous Driving. In *Proceedings of the 29th Annual International Conference on Mobile Computing and Networking*. ACM, 1–15. doi:10.1145/3570361.3613280
- [20] Yuankai He and Weisong Shi. 2024. A Vision for Transformative Intersections. *Computer* 57, 12 (2024), 58–68. doi:10.1109/MC.2024.3421963
- [21] Huali Intelligent Mobility. 2021. MOCAR OBU and HUALI RSU Receive LTE-V2X Protocol-Conformance Certification. https://www.hualitec.com/news_page.php?cid=24&id=287
- [22] International Telecommunication Union. 2023. *Framework and Overall Objectives of the Future Development of IMT for 2030 and Beyond*. Technical Report Recommendation ITU-R M.2160-0. International Telecommunication Union Radiocommunication Sector. <https://www.itu.int/rec/R-REC-M.2160-0-202311-I/en>
- [23] Grigorios Kakkavas, Maria Diamanti, Kwame Nseboah Nyarko, Matthias Gabriel, Vasileios Karyotis, Klaus Moessner, and Symeon Papavassiliou. 2022. Realistic Field Trial Evaluation of a Tele-Operated Support Service for Remote Driving over 5G. In *2022 IEEE Conference on Standards for Communications and Networking*. IEEE, 58–63. doi:10.1109/CSCN57023.2022.10051034
- [24] Yu-Jen Ku, Bryse Flowers, Samuel Thornton, Sabur Baidya, and Sujit Dey. 2022. Adaptive C-V2X Sidelink Communications for Vehicular Applications Beyond Safety Messages. In *2022 IEEE 95th Vehicular Technology Conference*. IEEE, 1–6. doi:10.1109/VTC2022-Spring54318.2022.9860580
- [25] Ruoshen Mo, Bo Wu, Zhaowei Tan, and Hang Qiu. 2025. SEE-V2X: C-V2X Direct Communication Dataset: An Application-Centric Approach. In *Proceedings of the 23rd ACM Conference on Embedded Networked Sensor Systems*. ACM, 305–311. doi:10.1145/3715014.3722077
- [26] Jad Nasreddine, Paul Salvati, and Miguel Fuentes. 2026. Accelerating the Uptake of 5G for Automotive: Real-World Trials from the TARGET-X Project. *Future Internet* 18, 4 (2026), 189. doi:10.3390/fi18040189
- [27] Next Generation Mobile Networks Alliance. 2021. *5G Time-Division Duplex Uplink*. Technical Report Version 1.0. Next Generation Mobile Networks Alliance. [18](https://www.ngmn.org/wp-content/uploads/220117-

</div>
<div data-bbox=)

5G-TDD-Uplink-White-Paper-v1.0.pdf

- [28] one6G Association. 2025. *6G Technology Overview* (5th ed.). Technical Report. one6G Association. <https://one6g.org/download/6324/>
- [29] Hang Qiu, Po-Han Huang, Nam Asavisanu, Xiaochen Liu, Konstantinos Psounis, and Ramesh Govindan. 2022. AutoCast: Scalable Infrastructure-Less Cooperative Perception for Distributed Collaborative Driving. In *Proceedings of the 20th Annual International Conference on Mobile Systems, Applications and Services*. ACM, 128–141. doi:10.1145/3498361.3538925
- [30] SAE International. 2020. *Vehicle-to-Everything Communications Message Set Dictionary*. SAE International. doi:10.4271/J2735_202007
- [31] SAE International. 2024. *Vehicle-to-Everything Communications Message Set Dictionary*. SAE International. doi:10.4271/J2735_202409
- [32] Shuyao Shi, Jiahe Cui, Zhehao Jiang, Zhenyu Yan, Guoliang Xing, Jianwei Niu, and Zhenchao Ouyang. 2022. VIPS: Real-Time Perception Fusion for Infrastructure-Assisted Autonomous Driving. In *Proceedings of the 28th Annual International Conference on Mobile Computing and Networking*. ACM, 133–146. doi:10.1145/3495243.3560539
- [33] Shuyao Shi, Neiwen Ling, Zhehao Jiang, Xuan Huang, Yuze He, Xiaoguang Zhao, Bufang Yang, Chen Bian, Jingfei Xia, Zhenyu Yan, Raymond W. Yeung, and Guoliang Xing. 2024. Soar: Design and Deployment of a Smart Roadside Infrastructure System for Autonomous Driving. In *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking*. ACM, 139–154. doi:10.1145/3636534.3649352
- [34] Eugene Song, Davide Pesavento, and Ya-Shian Li-Baboud. 2024. *C-V2X Interoperability Testing Datasets: Description and Use*. Technical Report NIST Technical Note 2315. National Institute of Standards and Technology. doi:10.6028/NIST.TN.2315
- [35] U.S. Department of Transportation. 2026. CARMA Platform System Description. Accessed August 12, 2026. https://usdot-fhwa-stol.github.io/documentation/carma-platform/md_docs_System_description.html
- [36] U.S. Department of Transportation. 2026. V2X Hub. Software repository. Accessed August 12, 2026. <https://github.com/usdot-fhwa-OPS/V2X-Hub>
- [37] Vehicle Safety Communications Consortium. 2005. *Vehicle Safety Communications Project: Task 3 Final Report*. Technical Report DOT HS 809 859. National Highway Traffic Safety Administration. <https://www.nhtsa.gov/document/vehicle-safety-communications-project-task-3-final-report>
- [38] Bo Wu, Ruoshen Mo, Justin Yue, Yanyu Zhang, Janice Nguyen, Guoyuan Wu, Amit Roy-Chowdhury, Matthew J. Barth, and Hang Qiu. 2026. CooperScene: Multi-Modal Cooperative Autonomy Benchmark with C-V2X Communication Characterization. In *European Conference on Computer Vision*. <https://cis.ucr.edu/CooperScene/>
- [39] Tianze Wu, Sa Wang, Yungang Bao, and Weisong Shi. 2024. Tentacles: A Middleware with Multi-Network Communication Reliability for Vehicle-Infrastructure Cooperative Autonomous Driving. In *2024 IEEE 100th Vehicular Technology Conference (VTC2024-Fall)*. IEEE, Piscataway, NJ, USA, 1–8. doi:10.1109/VTC2024-Fall63153.2024.10758047
- [40] Lu Zhan, Miriam Di Russo, Debashis Das, Kevin Stutenberg, Priyash Misra, Jongryeol Jeong, and Eunjeong Hyeon. 2026. Characterizing the Impact of C-V2X PC5 Radio on Cooperative Driving Automation in Hardware-in-the-Loop Experiments. In *SAE Technical Paper Series*. SAE International. doi:10.4271/2026-01-0077
- [41] Xumiao Zhang, Anlan Zhang, Jiachen Sun, Xiao Zhu, Y. Ethan Guo, Feng Qian, and Z. Morley Mao. 2021. EMP: Edge-Assisted Multi-Vehicle Perception. In *Proceedings of the 27th Annual International Conference on Mobile Computing and Networking*. ACM, 545–558. doi:10.1145/3447993.3483242

A IPI Reference Schema

Table 5 summarizes the public data model used by the reference implementation. Message-mode calls use an envelope around a typed or pre-encoded payload. Operation-mode calls add a session identifier and operation state. The reference facade stores active sessions and responses in memory and exposes registration, heartbeat, telemetry, invocation, response retrieval, and termination calls.

The C++17 implementation maps supported J2735 structures into the common IPI envelope. It preserves externally encoded J2735 messages byte for byte. The cooperative-service object uses a canonical binary encoding. Its validators bound identifiers, strings, waypoints, detected objects, control commands, confidence, and opaque payload length. Seven automated tests cover selected BSM and PSM round trips, a SPaT transport loopback, one cooperative-message round trip, byte preservation, truncated-input rejection, session operations, topic parsing, and TCP/MQTT probe loopback. Correlation-identifier uniqueness, state-transition policy, security, admission, scheduling, and path selection remain application and deployment policies in the reference interface.

B Deployment and Measurement Details

Table 6 expands the deployment context from Section 4. The AirSpeed inventory records versions 22.0-24.0.0 and 22.00-53.0.0. Its two enabled n48 cells map to two sector carriers. The primary application traffic used 70/20/10 and Cell 2. The August 15–19 TDD study records application direction, serving cell, location, reported signal context, and both cells’ administrative state. Seven overlapping DU Cell exports provide five-minute cell counters for the available experiment windows. Their zone-less timestamps are interpreted as EDT from traffic alignment. The CBRs export records Category B General Authorized Access operation, one requested and authorized grant, no suspended grant, and a successful Spectrum Access System heartbeat.

One CAV gateway serves as the sole physical experiment UE. The dedicated radio and 40-MHz channel operate in a clean n48 band without ambient contention. The load experiment places one to four streams behind it. The client-demand experiment places 1–100 closed-loop clients behind the same attachment. These workloads traverse the gateway, Uu path, private core, transport, and d1 responder. The matrix excludes public-network handover and roaming. This controlled radio-band context removes ambient users as an explanation for the results. Profile transitions can still change route, cell state, serving cell, signal, and transport state, so those fields remain part of each comparison.

The 5G radio survey comprises three continuous drives through the testbed. After removing 123 records from a

frozen radio-value interval, it retains 1,178 band-48 records with a median RSRP of -108 dBm and a median recovered NR SINR of 9 dB. We use this survey for route-level coverage and retain the signal records collected with each stationary application campaign for its workload comparisons. The PC5 collections include a reference point, a building-obstructed NLOS point, successive stationary field points, and four driven routes. Exact GNSS coordinates are transformed to a local frame in the analysis artifacts.

C Workload and Experiment Matrix

The PC5 stationary sweep sends 0, 0.25, 0.5, 1, and 2 KiB, normally for 1,000 requests per size at 100-ms intervals and a 1,000-ms harness timeout. Timeout-dominant collections retain every transmitted request. The four route collections send 0.25-KiB requests at 200-ms intervals. One stopped route contains 675 attempts; the other routes contain 1,000 attempts each.

The five stationary uplink-heavy 5G payload campaigns were collected on May 13, 14, 15, 21, and 22, 2026. Most completed conditions contain 1,000 requests at 200-ms intervals. Every application request uses an IPI cooperative-service frame. The requested size denotes its application content; sender logs retain the encoded IPI frame size. The dataset manifest contains 22,431 artifacts from OpenDAIR-V2X, V2X-Seq, TruckV2X, and V2X-Radar. Images span 25.5–666.9 KiB, point clouds span 15 KiB–3.85 MiB, and maps span 22.3–44.9 MiB. These ranges motivate the fixed-size sweeps. The 128–512-KiB and 1–2-MiB transfers represent chunks drawn from these application-payload scales.

We derive payload sizes from 922 saved V2X-Radar validation outputs. The model assigns a 256-B header and 96 B per predicted object. Payloads range from 12.9 to 24.4 KiB, with median, p95, and p99 sizes of 19.2, 22.3, and 23.4 KiB. Network replay fills an IPI cooperative-service content field to each representative size. The 58.6-KiB condition is a separate stress payload. The responder returns a compact correlated acknowledgment immediately after decoding the frame.

The August 15–16 TDD runs retain the same uplink-heavy timing contract. The August 17 matched experiment instead uses a compact vehicle request followed by an exact 1-, 10-, 100-, or 500-KiB edge-to-vehicle response. Each profile contains 1,000 validated TCP and MQTT exchanges per payload. The vehicle timer includes complete payload receipt and sequence, length, and checksum validation. The same experiment then performs ten exact 50-MiB TCP uploads and ten exact downloads per profile. Endpoint byte counts, checksums, and receiver-observed durations define application goodput.

The August 18 40/40/20 repeat uses 1,000 uplink-heavy and downlink-heavy exchanges at 1, 10, and 100 KiB. Each

Table 5: IPI reference data model.

Object	Principal fields	Function
Envelope metadata	Message, source, intersection, send time, transport class, optional session and correlation identifiers	Identifies an independent message or envelope-based operation
J2735 payload	Message type, encoding, typed content or pre-encoded bytes, optional frame counter	Carries typed BSM, PSM, MAP, SPaT, SRM, and SSM profiles or preserves any encoded J2735 message
CAV operation request	Operation class, vehicle, request, optional session, location, application context, desired horizon, opaque data	Describes a perception, planning, control, or map operation
CAV operation result	Session, vehicle, operation class, progress state, expiration, confidence, typed or opaque content	Represents an update, completion, or rejection associated with a CAV operation
Session records	Vehicle profile, role, requested operations, state, lease, heartbeat interval, telemetry, responses	Maintains the application state used by continuing operations

Table 6: Detailed radio, gateway, and endpoint configuration.

Element	Retained configuration	Measurement record
5G radio	Dedicated AirSpeed 2900; NR n48; TDD; 40 MHz; 30-kHz control and synchronization-signal subcarrier spacing; 20-ms synchronization block period	Clean n48 band without ambient traffic; one uplink and up to two downlink layers configured; downlink 256-quadrature-amplitude modulation enabled as a cell capability
Cells	Cell 1: absolute radio-frequency channel number 637992 and synchronization signal at 3558.720 MHz; Cell 2: 645334 and 3669.600 MHz	Primary traffic and the matched August 17 blocks used Cell 2; the second August 16 40/40/20 TCP repetition used Cell 1; the 70/20/10 Cell 2 lock was operator-reported
Power and antenna	33-dBm cell transmit power; 34 dBm/MHz effective isotropic radiated power; 17-dBi cross-polarized integrated antenna	Product pattern approximately 65° azimuth by 8° elevation; structural centerline 20 ft above roof level
Duplex configuration	Established application profile: 70/20/10 downlink/uplink/dynamic frames; August 15–19 comparison: 40/40/20 and 70/20/10; fixed 10D4G packing	Route, cell state, and serving cell changed during the August 15–16 transition; August 17 and 19 supplied matched workload comparisons; separate 60/20/20 runs are excluded; the attempted 30/60/10 configuration produced no valid experiment
Vehicle gateway	Cisco Meraki MG52-HW campaign gateway with Telit FN990A40 modem and internal antenna; GLiNet GL-X3000 replacement gateway; 1-Gbit/s full-duplex Ethernet	The August 19 GL-X3000 repeat records gateway RSRP, RSRQ, and SINR while evaluating both TDD profiles at one stationary location
Session and endpoint	SIM provisioned on Data Network Name cisco5g; operator-reported default 5G QoS Identifier 9; on-site MX250 and directly connected d1 host	Host captures observed TOS 0x0 for default and 5qi-mapped experiment labels
Field context	Sole experiment UE; separate three-drive survey with a Samsung 22 handset running G-NetTrack Pro; 1,178 valid band-48 records; RSRP –121 to –88 dBm; recovered NR SINR –20 to 30 dB	Drive-survey medians: –108 dBm RSRP and 9 dB SINR; August 17: operator-reported –100-dBm RSRP and –13-dB RSRQ; August 18: –105 to –115 dBm; August 19 gateway medians: –99 and –98 dBm RSRP and 20-dB SINR
PC5 hardware	Mocar OBU, HUALI RSU, 2020 vendor J2735 stack, packet type 0x1b, 4,080-B installed-interface limit	GNSS-joined application outcomes; no received-power, channel, resource-pool, modulation, or retransmission export

transport contains 250 declared uplink-heavy exchanges at 500 KiB and 500 downlink-heavy exchanges. Reported RSRP changes from –105 to –110 and then –115 dBm during the run. Its shortened bulk stage contains one complete upload/download pair and one additional upload. The failed August 17 downlink preflight, the bulk pilot, and the unintended August 18 partial preflight remain diagnostic artifacts outside the primary estimates. The August 19 GL-X3000 repeat uses 500 validated exchanges for every combination of direction, transport, profile, and 1-, 10-, 100-, or 500-KiB payload. It adds three exact 50-MiB uploads and three exact downloads per profile. The two profile blocks retain separate gateway radio records and use the same stationary location. Table 7 records the controlled conditions and the corresponding measurement for every experiment family.

In the closed-loop application-demand collection, each application client waits for a response or timeout before the

next request from that client and then applies a 200-ms interval. At negligible response time, five requests/s/client is the maximum nominal rate. A locked-cell 70/20/10 collection additionally uses four conditions: one idle 1-KiB client, one idle 23.4-KiB client, the 1-KiB client with offered uplink traffic, and 100 1-KiB clients. TCP, MQTT, and UDP occupy separate five-minute collection bins with two repetitions. The first three conditions use 500 probes per repetition; the 100-client condition uses 500 probes per client per repetition.

Every issued primary request/response attempt remains in its condition denominator, including missing, timed-out, late, and failed attempts. Entirely excluded diagnostic roots and never-started bulk repetitions do not enter a primary estimate. Operator-shortened conditions use their declared retained counts. The request/response experiments carry application-derived workloads in IPI cooperative-service frames and measure RTT and deadline completion. PC5 uses

Table 7: Complete workload and communication experiment matrix.

Question	Controlled conditions	Recorded outcome
Message and compact exchange	J2735 callback checks; local IPI tests; PC5 echo; Uu 0–4-KiB IPI frames over TCP/MQTT	Decode or callback result, response completion, RTT
Payload scale	PC5 0–2 KiB; uplink-heavy Uu IPI frames with 0–2 MiB vehicle-originated content; downlink-heavy Uu frames with 1–500 KiB returned content; detector-derived sizes; 58.6-KiB stress payload	$C(D)$ and response p50/p95/p99
Field and route coverage	PC5 stationary points and four routes; original Uu stationary collections and Samsung 22 continuous-drive survey; matched August 17 location 3; August 18 time-segmented context; August 19 same-location gateway measurements	GNSS-joined response completion, RTT, and separate radio context
Protocol formation	TCP, MQTT QoS 0 over TCP, raw UDP, and UDP with 1,400-B application fragments	Returned acknowledgment after decode or returned payload after reassembly and validation
TDD and application direction	August 15–16 uplink-heavy repetitions; August 17 matched downlink-heavy responses and exact transfers; August 18 weak-signal direction repeat; August 19 same-device 1–500-KiB bidirectional and exact-transfer matrix under both profiles	Profile-transition repeatability, response RTT, deadline completion, exact application goodput, device sensitivity, and radio context
Exact directional transfer	Ten exact 50-MiB uploads and downloads per August 17 profile; three per August 19 profile and direction; receiver byte-count and checksum validation	Individual transfer goodput, mean, and 95% confidence interval
Competing traffic	Separate historical 15-minute offered controls; 1-KiB foreground with one to four offered 25-Mbit/s uplink streams	Achieved background rate and foreground deadline completion
Application demand	1, 2, 5, 10, 20, 50, and 100 clients; 1-KiB closed-loop requests; sole experiment UE	Aggregate and per-client response completion and RTT
Packet marking	Default and 5qi-mapped application labels	Vehicle-interface Internet Protocol header capture
Interruption	TCP receiver, UDP receiver, MQTT receiver, or MQTT broker restarted for 10 s	Missing attempts, deadline completion, and response gap

sequence-matched workload echoes. Exact bulk transfers instead use validated bytes, checksum, receiver duration, and application goodput.

D Detailed Results for Additional Conditions

Several experiment families include conditions that are not represented by dedicated figure panels. This section reports their complete numerical outcomes.

D.1 Field-Specific Uplink Payload and Transport Results

Field collections changed the response tail across both payload and transport. In one collection, 0–4-KiB p95 RTT was 36–49 ms for MQTT and 100–140 ms for TCP. At 2 MiB, p95 reached 2,536 and 3,633 ms. At another location, 2-MiB TCP p95 reached 10,767 ms. In a variable-path collection, empty-payload TCP and MQTT p95 rose to 1,142 and 510 ms. The 256-KiB MQTT condition returned 999/1,000 responses with a 43,836-ms p95.

The transport comparison used empty-payload controls, detector-derived payloads, and the 58.6-KiB stress payload. In the –106-dBm collection, TCP and MQTT returned every response at each size. MQTT p95 RTT increased from 49.826 ms with an empty payload to 183.705 ms for the 23–24-KiB group and 403.790 ms at 58.6 KiB. The corresponding TCP values were 151.696, 169.540–213.894, and 381.409 ms. In the –120-dBm collection, 58.6-KiB MQTT and TCP p95/p99 increased to 770/1,230 and 662/1,027 ms.

Raw UDP returned 1,000/1,000 empty-payload and 300/300 1-KiB responses. It returned 0/45, 0/100, and 0/100 at 1.37, 4, and 19.2 KiB. Application fragmentation and receiver reassembly increased response completion at –106 dBm. Completed responses reached 99.5–100.0% through 23.4 KiB, while the 58.6-KiB condition returned 20/100 responses. At –120 dBm, payloads through 4 KiB returned 99.9–100.0%, 19–25-KiB payloads returned 93.5–96.1%, and 58.6 KiB returned 0/100.

D.2 Gateway and TDD Replication Results

At 70/20/10, the closest MG52 control and the GL-X3000 repeat had reported or measured median RSRP of –100 and –99 dBm. The GL-X3000 repeat measured 15.691 Mbit/s upload and 166.106 Mbit/s download goodput. These rates are 30.8% and 31.0% above the MG52 control rates of 12.000 and 126.786 Mbit/s. Under 40/40/20, the earlier MG52 collection crossed weaker reported signal levels from –105 to –115 dBm. The later GL-X3000 collection at median –98 dBm measured 10–500-KiB vehicle-to-edge p50 RTTs 63.7–92.4% below the earlier MG52 collection, so the earlier large-uplink collapse did not recur. The same-device comparison still produced lower burst latency and higher sustained goodput under 70/20/10. The higher rates in the replacement-device run therefore did not remove the measured gap between the two deployed configurations.

Consecutive 70/20/10 runs changed TCP p95 RTT by only 0.22–1.29% across 1–100 KiB. During the initial 40/40/20 rollout, the route required recovery, cell administrative state changed, and separate repetitions selected different serving

cells. A later matched MG52 downlink block completed all 8,000 exchanges, although its payload and transport effects differed. The GL-X3000 repeat completed both directional matrices but retained the large-object latency and goodput gap.

D.3 Sustained-Stream Load and Packet-Marking Results

The two -106 -dBm repetitions achieved 1.320–2.267 Mbit/s of background traffic. MQTT $C(100\text{ ms})$ fell from 99.70% at idle to 40.15% with the generator active. At 200 ms, it changed from 99.95% to 98.00%. TCP achieved 0.75% and 0.40% at 100 ms. Its 200-ms completion fell from 97.90% to 59.00%. Across one, two, and four background streams, achieved throughput ranged from 1.990 to 5.743 Mbit/s. The pooled active-generator conditions in two weak-band collections produced MQTT/TCP 200-ms completion of 83.63%/38.40% and 91.90%/49.17%.

At -106 dBm, the active generator increased MQTT p95 from 52–60 to 174–176 ms. TCP p95 increased from 162–194 to 388–420 ms. Across the one-, two-, and four-stream sweep, MQTT p95 was 170, 468, and 182 ms. TCP p95 was 391, 402, and 410 ms. The MQTT tail did not vary monotonically with the number of streams.

The default and 5qi-mapped labels both produced observed TOS values of 0×0 in the vehicle capture. The foreground and background flows entered the measured path without an observed packet-class difference between the requested labels.

D.4 Concurrent-Client Results

In the -106 -dBm seven-point repeat, 100 clients produced aggregate p95 RTTs of 81.850 ms for MQTT, 264.070 ms for TCP, and 113.391 ms for UDP. MQTT and TCP each returned 100,000 of 100,000 responses. UDP returned 99,965 of 100,000. A separate 100-client repeat yielded p95 values of 177.053, 283.765, and 145.894 ms and returned 100,000, 100,000, and 99,365 responses.

The 100-client collection at -120 dBm produced a different ordering. MQTT and TCP p95 rose to 5,135.984 and 2,908.420 ms. They completed 99.499% and 100% of issued requests before the harness timeout. UDP completed 90,811 of 100,000 requests, with a p95 of 172.500 ms. Its shorter response tail coincides with 9,189 missing cycles.

The per-client distributions reveal substantial unfairness within the shared attachment. Across 100 clients, the median client p95/p99 was 3,814.102/13,149.421 ms for MQTT and 2,938.408/6,400.146 ms for TCP. Many flows fall outside the 100- and 200-ms application budgets even though most requests still receive a response before the harness timeout.